COLLISION RECORDS, ARITHMETIC, AND RAW LOCAL INVERSION IN A TRIANGULAR LORENTZ GAS

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Abstract. We study the exact-label raw pointwise local limit problem for actual returns in a triangular Lorentz gas, with the original section, guard, arithmetic reference and order of limits. We retain the optical trace and positive variation payments, global roof–angle uniqueness across disconnected physical components, and the complete target-roof marked pressure sum. The latter now gives an exponential moment for the joint depth and logarithmic severity of all original incidence witnesses, without an independence assumption. Optimizing this moment pays the source of load at least $(2+\eta)\log(m+1)$ in the m^2-normalized essential-height topology. The actual two-end depth polynomial also gives a quadratic witness-count tail; the number of witnesses needed for a normalized payment is of order $\sqrt{\log m}$, rather than $\log m/\log\log m$. In particular sufficiently severe shallow incidences are paid without requiring a deep collision or a large witness count. All source coefficients remain in an exact positive identity with the unchanged nongrazing comparison pool. The complete raw pointwise assertion remains the stated low-load-shortage limit; low-load shallow incidences and arbitrary-depth clearance are not declared small by the unscaled pressure theorem.

Introduction

The proved results of this article are global roof–angle uniqueness, target-roof marked pressure relative to the explicit geometric inputs, and collision-normalized positive payments for deep, numerous and high-load incidence sources. The complete exact-label raw pointwise local limit assertion (1.5) is still the research target: its equivalent remaining estimate (14.8) is not proved here. The new load and quadratic-count theorems strengthen the positive source estimates within that same problem, not a replacement problem.

The pointwise raw local limit problem concerns the density at the original target roof, uniformly in collision count, radius and exact return labels. Control of a fixed-window average does not by itself control that density. For the original positive microscopic paired source $\mathsf{U}_B^{\mathrm{micro}}$ and its full nongrazing comparison pool $P_{r_B}^{\mathrm{ng}}$, the estimate left by the preceding argument is

$$\lim_{B\to\infty}\limsup_{m\to\infty}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}_{B,m}^{\mathrm{micro}}(t)-K_B P_{r_B,m}^{\mathrm{ng}}(t))_+ = 0. \tag{0.1}$$

This is equivalent, within the retained scalar reduction, to the original pointwise assertion (1.5). It is not yet proved here. The retained source-side contribution is a quantitative payment of a further positive part of this same source and a physical description of every coefficient which survives that payment.

The distinction between source and capacity is essential. On a segment with an actually visited grazing collision at one end, the original paired minimum is zero. Its endpoint maximum, and hence its comparison capacity, need not be zero. The optical and trace arguments retained in Sections 3–6 pay this entire visited-grazing capacity without deleting unvisited tangencies or

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positive critical and section-boundary traces. They start from the actual broken-path action, use a positive continuant denominator, extend its reduced Hessian continuously to visited grazing, and integrate the inverse gradient along a fixed Gauss angle. All these arguments and their physical hypotheses are retained in the present article.

The source-side argument uses the total variation of the weighted density on an entire physical segment. If c^{ε} is its endpoint minimum and $\mathcal{V}^{\varepsilon}$ its variation, we split, before any roof pushforward,

$$c^{\varepsilon} = \min\{c^{\varepsilon}, \gamma_B \mathcal{V}^{\varepsilon}\} + (c^{\varepsilon} - \gamma_B \mathcal{V}^{\varepsilon})_+, \qquad \gamma_B = \max\{2, \varepsilon(B)^{-1/32}\}.$$

The first term can be positive, even when both endpoint traces are positive. Theorem 7.2 pays its complete physical sum by the original uncut weighted volume, with no residence lower bound, word count or comparison reach assumption. The resulting ordered height is $O(B^{-95/384})$.

For the second term, Theorem 8.1 shows that the density stays within relative distance γ_B^{-1} of its endpoint minimum and the original guard stays positive at every point of the closed segment. In particular, a single point where the original guard vanishes pays the entire original paired coefficient. This assertion is independent of the name of either endpoint mechanism. It does not assert that a single incidence or clearance test witnesses the guard throughout the segment.

Theorem 8.3 retains the full nongrazing comparison pool and gives an exact three-term positive identity for the original microscopic source. The remaining estimate is printed in (8.9), in the original essential-height topology and the original order of limits. Neither nearly constant density on a short segment nor finite-count exhaustion is identified with a proof of this remaining estimate.

Appendix B reconstructs the signed corner cofactor and proves a simultaneous multi-mark expansion. It also states the simultaneous one-sided endpoint lemma separately and makes the uniform constant in the trace modulus explicit. The multi-mark bound is per trajectory; it is not a new weighted-pressure theorem. The complete preceding article and historical derivations are preserved byte-for-byte in the accompanying archive [2]. The physical pairing model, regular derivative estimate and spectral average estimates remain source-pinned inputs rather than consequences of finite algebraic diagnostics.

From momentum fibers to the original target roof. A geometric pressure bound on endpoint momentum fibers does not by itself control a target-roof fiber: the momentum magnitudes of its words need not agree. Theorem 9.2 removes this obstruction to summation. Convexity of the Cartesian broken-path length gives two support inequalities between any two physical trajectories of one lifted itinerary. On a common endpoint-gradient ray these inequalities order their lengths strictly by gradient magnitude. Thus a word supplies at most one point at a specified roof and Gauss angle, without requiring a connected word domain.

Theorem 10.3 combines this global fact with the retained uniform geometric pressure and optical stable-cocycle comparison. For any set S of distinct interior marks it gives

$$\left\| \frac{dF_{\#}(\prod_{i\in S} c_i^{-1} \mathbf{1}_{\Pi} \nu_R^*)}{dt} \right\|_{\infty} \le C_\star K_\star^{|S|}.$$

All original exact labels remain on the left. On the upper side, a marked full word is reconstructed from its unmarked blocks and its finitely many deleted physical step symbols. This is a weighted summation theorem; it does not insert an uncontrolled count of words or homogeneous strips. Its constants have no Gaussian m^{-2} factor, a distinction essential to its use.

Corollary 10.4 sums changing incidence witnesses by elementary symmetric polynomials. The original geometric depth thresholds have bounded sum, independently of m. Theorem 11.2 consequently pays every persistent-source trajectory with an incidence witness beyond an explicit $O(\log m)$ end layer, with normalized height $C\varepsilon(B)(m+1)^{-\eta}$. A factorial bound also pays states

with $O(\log m/\log\log m)$ or more incidence witnesses. These collision-dependent classifications are substituted only into bounds valid at every finite count; all spectral windows and outer scales remain fixed in their original order.

This leaves the original pointwise target, not a new topic, at (11.12). The remaining shallow incidence and clearance source is not declared small by an unscaled pressure estimate. Sections 9–11 contain the new proofs; all preceding source and capacity arguments remain in the article, with the complete historical packet retained in `archive/v111/archive/v110` [2].

Joint incidence load and the two collision-end layers. A witness at depth d_i already spends the amount αd_i of logarithmic incidence load, where $\alpha = |\log q_0|/4$. Its additional smallness spends $\log(2\varepsilon q_0^{d_i/4}/c_i)$. These contributions are nonnegative and can be summed before selecting a witness. The resulting variable $\mathscr{L}_\varepsilon$ obeys

$$\left\| \frac{dF_{\#}(e^{\lambda\mathscr{L}_\varepsilon}\mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \le C_\star \exp\left\{ \frac{4K_\star\varepsilon}{e^{\alpha(1-\lambda)}-1} \right\}, \qquad 0<\lambda<1.$$

Theorem 12.1 proves this by a positive product expansion on the original physical source. Optimization gives the tail $C_\star \exp[-u+2\sqrt{b_\varepsilon u}]$, where $b_\varepsilon = 4K_\star\varepsilon/\alpha$. At $u=(2+\eta)\log(m+1)$ this pays the normalized height, including states with only one sufficiently severe shallow witness.

The depth multiset has at most two entries at every level. Lemma 13.1 preserves this fact in the coefficient of $\prod_{d\ge1}(1+zr^d)^2$, instead of bounding that coefficient by a factorial involving only the total threshold sum. It proves the quadratic exponent $r^{\lfloor (s+1)^2/4\rfloor}$. The resulting Theorem 13.2 reduces the witness-count threshold to order $\sqrt{\log m}$. Sharpness is established for this depth polynomial, not claimed for an unproved physical lower bound.

Theorem 14.1 incorporates the new payment as a positive restriction of the original source. Its retained part has bounded total incidence depth and bounded total logarithmic severity at the displayed collision scale. Clearance witnesses, including unvisited tangency, remain distinct and may occur at every depth. The full preceding article and all of its historical material are preserved in `archive/v111` [1]. The current article retains every preceding active proof, promotes the geometric input conventions explicitly, and supplies the unbounded-weight exhaustion and marked-step indexing used by the target-roof sum.

1. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0)+\mathbb{Z}(1/2,\sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha,p)\in(\mathbb{R}/2\pi\mathbb{Z})\times(-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\,\mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6,-1/2), \quad (-\theta_R,\sin\theta_R), \quad (\pi/3-\theta_R,\sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1-6q^2+q^4) = 0.$$

Set

$$U = \{|\alpha-\pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{1.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k, m, t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m - j)$ and $d'_j = \min(j, m - 1 - j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d'_j/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{1.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_{\#}(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{1.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

1.1. **The arithmetic reference and the target.** For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R, \vartheta)$ and $\mathsf{P}_j(R, \vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log \lambda_j(R, a_j), \quad H_j = -\nabla^2 \log \lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \ge a I_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R \nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} H v/2}\,\mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{1.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\le n\le m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{1.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2. The physical source and the inherited analytic inputs

We specify the inputs to the new geometric argument, separating the existence of the original physical source from the endpoint regularity that will be proved below. The complete historical proofs and their qualifications are retained in the accompanying archive.

For fixed m, R, n, k, let Θ_{phys} be the original disjoint incident-component pairing-sheet space. It is the countable Borel sum of the maximal regular components of the original exact event, parametrized by

$$(u,\theta) = (F, \arg \nabla_q F), \qquad q = (\alpha_0, \alpha_m), \qquad a(\theta) < u < b(\theta), \quad \ell = b - a > 0.$$

The sheet identity, and not just its angular coordinate, is included in θ. Integration $d\theta$ means the original one-copy physical sum with angular Lebesgue measure. The regular components, their Borel endpoint maps, their physical multiplicity convention, and their common coarea versions are the retained pairing-space input. This article does not replace them by symbolic collision words. Every component carries the same exact event (1.1). Put

$$\begin{aligned} &g = \nabla_q F = R(-p_0, p_m), \quad z = |g|, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i, \\ &\omega = |\partial_{p_0} \alpha_m|^{-1}, \quad \chi = |\det D_q(F, \arg g)|, \quad A = \frac{\omega}{4\pi c_* \chi}, \quad f^\varepsilon = AW^\varepsilon. \end{aligned} \tag{2.1}$$

The density formula is used initially on the regular interior. Once its endpoint traces have been established, the original pairing is

$$f_-^\varepsilon = f^\varepsilon(a+,\theta), \quad f_+^\varepsilon = f^\varepsilon(b-,\theta), \quad c^\varepsilon = \min(f_-^\varepsilon, f_+^\varepsilon), \quad d\pi^{\varepsilon,+} = c^\varepsilon \mathbf{1}_{\{a<u<b\}} d\theta\, du. \tag{2.2}$$

Write $p_\vee = \max(f_-^\varepsilon, f_+^\varepsilon)$ and $e_\partial = |f_-^\varepsilon - f_+^\varepsilon|$. Thus $p_\vee = c^\varepsilon + e_\partial$.

Input statement 2.1 (Regular variation and completed physical averages)**.** The inherited regular-point optical calculation gives, almost everywhere on each regular segment,

$$|\partial_u f^\varepsilon| \le C(1 + \varepsilon^{-1}) A I_m / z. \tag{2.3}$$

Let $\mathsf{T}_\Delta^\varepsilon$ be the complete paired roof density restricted to $0 < \ell < \Delta$, with no additional selector, and let

$$v^0(t)\, dt = F_\#(I_m z^{-1} \mathbf{1}_\Pi \nu_R^*), \qquad A_r f(t) = (2r)^{-1} \int_{t-r}^{t+r} f(u)\, du.$$

For every previously fixed $\varepsilon, r, \Delta, \rho > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r \mathsf{T}_\Delta^\varepsilon\|_\infty \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}, \tag{2.4}$$

$$\limsup_m \sup_{R,n,k} m^2 \|A_\rho v^0\|_\infty \le C. \tag{2.5}$$

The constants are independent of the previously fixed widths. The coarea identity identifies the full physical sum of AI_m/z with v^0. Fixed-count local finiteness and common roof versions are retained. These are averaged, not pointwise, bounds.

These are the positive pairing, uncut volume and weighted optical variation statements in Section 2 of the frozen article [4]. Their exact historical source labels are recorded in the proof ledger. The new proof uses (2.3) only on regular interiors and supplies its integrable passage to the physical endpoints.

2.1. **The original microscopic source.** The outer schedule is

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad r_0 = c_0^{\mathrm{out}} \varepsilon(B)^4, \qquad \delta(B) = \tfrac{3}{4} r_0, \qquad \Delta_B = 8\delta(B). \tag{2.6}$$

The fixed constants A_0, c_0^{out} are the inherited constants; c_0^{out} is written this way to avoid confusion with the initial incidence c_0. Keep the original remainder selector H_B, the paid critical-arm family $\mathscr{C}_B$, and the adaptive paid family $\mathscr{D}_B$ of the historical microscopic reduction. Their memberships are Borel, disjointized by first physical membership, and their total outer costs are at most $C\varepsilon(B)^{-1/32}$. For a previously fixed $0 < h_B < \delta(B)$ put

$$\begin{aligned} \widetilde{H}_B &= H_B \mathbf{1}_{(\mathscr{C}_B \cup \mathscr{D}_B)^c} \mathbf{1}_{\{0<\ell<h_B\}}, \\ \mathsf{U}_B^{\mathrm{micro}}(t) &= \int c^{\varepsilon(B)} \mathbf{1}_{\{a<t<b\}} \widetilde{H}_B(\Psi(\theta,t))\, d\theta. \end{aligned} \tag{2.7}$$

No sign cancellation, deletion of an endpoint class, or change of section occurs in this definition. Here Ψ denotes the original physical parametrization.

Input statement 2.2 (The scalar microscopic reduction)**.** For every fixed M, define

$$\mathcal{H}_M(T) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} T(t).$$

The historical exact positive reduction gives

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) + C_M B^{-1/384}. \tag{2.8}$$

The assertion (1.5) is equivalent to $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \to 0$ for every fixed M. All widths and costs are fixed at the outer stage before the complete physical sum, the essential suprema, and the collision limsup; B is outermost.

This is the unchanged theorem in historical module 255. The fixed-positive-residence estimate in module 253 and the earlier paid-arm estimates are retained as its inputs. A bound uniform in a previously fixed positive residence is not used at a collision-dependent residence.

3. The reduced optical action

Fix an actual collision itinerary with $m \ge 1$ flights. Its visited centres are $O_0, \dots, O_m$, its collision points are $x_i = O_i + Rn(\alpha_i)$, and $n(\alpha) = (\cos\alpha, \sin\alpha)$. Write $\tau_i = |x_{i+1} - x_i|$. All variations in this section are made on this actual itinerary before a physical boundary is reached. They neither continue an event through a forbidden collision nor change an occupation label.

Lemma 3.1 (The original section stays away from endpoint grazing)**.** *On the closure of the original initial and terminal section tests,*

$$|p_0|, |p_m| \le \frac{9}{10}, \qquad c_0, c_m \ge c_\circ := \frac{\sqrt{19}}{10}. \tag{3.1}$$

Proof. The only section centres with tangential coordinate exceeding $1/2$ in absolute value have coordinate $2q_R/(1+q_R^2)$. This function is increasing on $(0,1)$, and

$$\frac{2(53/100)}{1+(53/100)^2} + \frac{1}{200} < \frac{9}{10}.$$

The other squares and the rectangle U satisfy the same strict bound. Passing to their closures proves the assertion. □

Let $S(\alpha_0, \dots, \alpha_m) = \sum_{i=0}^{m-1} \tau_i$ be the broken-path action. At a regular specular trajectory its interior variables are stationary. Eliminating them locally gives the actual endpoint action $F(q)$, where $q = (\alpha_0, \alpha_m)$.

Lemma 3.2 (Second variation and its Schur complement)**.** *For a real variation $\xi = (\xi_0, \dots, \xi_m)$ at an actual regular trajectory,*

$$D^2S[\xi,\xi] = R\left(c_0\xi_0^2 + c_m\xi_m^2 + 2\sum_{i=1}^{m-1} c_i\xi_i^2\right) + R^2\sum_{i=0}^{m-1}\frac{(c_i\xi_i + c_{i+1}\xi_{i+1})^2}{\tau_i}. \tag{3.2}$$

The reduced Hessian $G = D_q^2F$ satisfies

$$R\,\mathrm{diag}(c_0, c_m) \le G, \qquad \lambda_\circ I_2 \le G \le \Lambda_\circ I_2, \quad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}. \tag{3.3}$$

These inequalities are uniform in the radius, the collision count, and the original exact labels.

Proof. For a flight with unit direction v, the Hessian of its length in its endpoints is the squared perpendicular displacement divided by the flight length. The acceleration of $Rn(\alpha)$ is $-Rn(\alpha)$. At the departure and arrival points, respectively, $v\cdot n_i = c_i$ and $v\cdot n_{i+1} = -c_{i+1}$. Also $(Rn_i')\cdot v^\perp = Rc_i$ and $(Rn_{i+1}')\cdot v^\perp = -Rc_{i+1}$. Thus one flight contributes $R(c_i\xi_i^2 + c_{i+1}\xi_{i+1}^2)$ and the corresponding square in (3.2). Summing proves that formula.

Let $H = D^2S$, $E = \{0, m\}$ and $I = \{1, \dots, m-1\}$. For $m > 1$, the interior block is positive definite. The implicit function theorem and differentiation of the stationary action give

$$G = H_{EE} - H_{EI}H_{II}^{-1}H_{IE}, \qquad \eta^{\mathsf{T}}G\eta = \min_{\zeta}(\eta,\zeta)^{\mathsf{T}}H(\eta,\zeta). \tag{3.4}$$

The order in the last expression places endpoint coordinates first. The diagonal endpoint terms in (3.2) prove the first lower bound. Choosing the interior variation to be zero gives $G \le H_{EE}$. The latter block is bounded above by $\{R_+ + 2R_+^2/\tau_-\}I_2 = (47/6)I_2$; the same bound covers the single-flight case, where there is no Schur complement. Lemma 3.1 proves the remaining lower bound. □

3.1. **A positive optical denominator.** Put

$$\mathsf{S}(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{C}(\kappa) = \begin{pmatrix} 1 & 0 \\ \kappa & 1 \end{pmatrix}, \qquad \kappa_i = \frac{2}{Rc_i}. \tag{3.5}$$

Let M_m be the product from the initial collision to the terminal collision, with the rightmost factor acting first:

$$\mathsf{M}_m = \mathsf{S}(\tau_{m-1})\mathsf{C}(\kappa_{m-1})\cdots\mathsf{S}(\tau_1)\mathsf{C}(\kappa_1)\mathsf{S}(\tau_0), \qquad \mathfrak{b}_m = (\mathsf{M}_m)_{12}. \tag{3.6}$$

For $m = 1$ the product is just $\mathsf{S}(\tau_0)$.

Lemma 3.3 (Exact denominator and a marked incidence bound)**.** *The actual endpoint-coordinate density is*

$$\omega = |\partial_{p_0}\alpha_m|^{-1} = \frac{|G_{0m}|}{R} = \frac{Rc_0c_m}{\mathfrak{b}_m}. \tag{3.7}$$

With $\varrho = 47/59$, one has

$$\omega \le \frac{47}{6}\varrho^{m-1}, \qquad \frac{\omega}{c_j} \le \frac{2209}{72}\varrho^{m-2} \quad (1 \le j \le m-1). \tag{3.8}$$

The second inequality is asserted when $m \ge 2$.

Proof. The first equality with G follows by differentiating $p_0 = -\partial_{\alpha_0} F/R$ with α_0 fixed. For the denominator, the interior block of H is $R^2 C_I J C_I$, where $C_I = \mathrm{diag}(c_1, \dots, c_{m-1})$ and J is tridiagonal with

$$J_{ii} = \kappa_i + \tau_{i-1}^{-1} + \tau_i^{-1}, \qquad J_{i,i+1} = \tau_i^{-1}.$$

The corner cofactor in (3.4) gives $|G_{0m}| = R^2 c_0 c_m / (\prod_i \tau_i \det J)$. Expansion of successive tridiagonal determinants gives $\prod_i \tau_i \det J = \mathfrak{b}_m$. This identity also follows by multiplying the second column in (3.6); both recursions start with $\mathfrak{b}_1 = \tau_0$. The empty determinant convention handles $m = 1$.

All entries in the optical factors are nonnegative. Multiplication of a positive second column $(b, d)^{\mathsf{T}}$ by $\mathsf{S}(\tau_i)\mathsf{C}(\kappa_i)$ changes its top entry to $(1+\tau_i\kappa_i)b + \tau_i d$. Since $1+\tau_i\kappa_i \geq 1 + 2\tau_-/R_+ = 59/47$,

$$\mathfrak{b}_m \geq \tau_- \varrho^{-(m-1)}. \tag{3.9}$$

For a fixed interior collision j, split the product immediately before and after $\mathsf{C}(\kappa_j)$. Its $(1,2)$ entry is an expression with nonnegative coefficients in κ_j, whose linear coefficient is $\mathfrak{b}_{\mathrm{right}}\mathfrak{b}_{\mathrm{left}}$. The two subproducts have respectively $m - j$ and j flights. Hence

$$\mathfrak{b}_m \geq \kappa_j \tau_-^2 \varrho^{-(m-2)}.$$

Substitution into (3.7), and $c_0, c_m \leq 1$, proves (3.8). □

Corollary 3.4 (The full inherited incidence insertion)**.** *For the unchanged insertion* $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$, *with* $q_0 = 47/53$,

$$\omega I_m \leq 500 \varrho^{(m-2)_+}. \tag{3.10}$$

This is an estimate on each actual physical trajectory; it involves no lower cutoff on any interior incidence.

Proof. For $m \geq 2$, $\sum_{i=1}^{m-1} q_0^{d_i} \leq 2q_0/(1-q_0) = 47/3$. Use (3.8) and $(47/6)\varrho + (2209/72)(47/3) < 500$. For $m = 1$ the sum is empty and the first optical bound suffices. □

No estimate for a sum over collision words is obtained by simply summing the right side of (3.10). In particular, this per-trajectory exponential estimate is not being substituted for a collision-uniform local limit theorem.

4. Completion of the original roofwise endpoint traces

We use the original incident-component pairing sheets and their constant-angle coordinates $(u, \theta) = (F, \arg \nabla F)$. On one of their positive-length maximal regular segments, write $a < u < b$, $g = \nabla F = z e(\theta)$ and $e^{\perp}(\theta) = (-\sin\theta, \cos\theta)$. The physical section and all original exact-event tests are retained.

Lemma 4.1 (Finite Gauss-coordinate length)**.** *Along a fixed-*θ *segment,*

$$\frac{dq}{dz} = G^{-1} e, \qquad \frac{du}{dz} = z e^{\mathsf{T}} G^{-1} e, \qquad 2\lambda_\circ \leq \frac{d(z^2)}{du} \leq 2\Lambda_\circ. \tag{4.1}$$

Consequently z *is strictly increasing, both endpoint limits of* q *exist, and*

$$\int_a^b \frac{du}{z(u,\theta)} \leq \lambda_\circ^{-1}(z_b - z_a) \leq \lambda_\circ^{-1} \min\{\sqrt{2} R_+, \sqrt{2\Lambda_\circ (b-a)}\}. \tag{4.2}$$

A critical endpoint of a positive-length segment can only be its incoming endpoint.

Proof. Differentiate $\nabla F(q) = ze$ and then $F(q) = u$. The bounds in (3.3) give (4.1). They also give $|dq/dz| \le \lambda_\circ^{-1}$. Since $0 < z \le \sqrt{2}R_+$, the two endpoint limits of q exist in the chosen angular lifts. Integrating $du/z = e^{\mathsf{T}} G^{-1} e\, dz$ proves the first inequality in (4.2). The second uses $z_b^2 - z_a^2 \le 2\Lambda_\circ(b-a)$ and $(z_b - z_a)^2 \le z_b^2 - z_a^2$. If $z_b = 0$, strict increase would contradict $z > 0$ in the interior. □

Lemma 4.2 (A continuous reduced Hessian at visited grazing)**.** *The reduced Hessian has a continuous extension along a fixed itinerary when any collection of its interior c_i tends to zero, provided $\tau_i \ge \tau_-$ and* (3.1) *hold. Its extension still satisfies* (3.3)*. If at least one interior c_j vanishes, the extended mixed entry G_{0m} vanishes.*

Proof. Let $L = \sum_{i=0}^{m-1} \tau_i^{-1}(e_i + e_{i+1})(e_i + e_{i+1})^{\mathsf{T}}$, and set $C_E = \operatorname{diag}(c_0, c_m)$. For $m > 1$ put

$$Z = 2I + RC_I^{1/2} L_{II} C_I^{1/2} \ge 2I.$$

Formula (3.4) can be written without inverse incidences:

$$\begin{aligned} G = {} & RC_E + R^2 C_E L_{EE} C_E \\ & - R^3 C_E L_{EI} C_I^{1/2} Z^{-1} C_I^{1/2} L_{IE} C_E. \end{aligned} \tag{4.3}$$

Every term is continuous for $c_i \ge 0$, since Z^{-1} remains bounded. The order inequalities pass to this limit. The marked bound in (3.8), together with $|G_{0m}| = R\omega$, proves the last assertion. There is no interior extension to make when $m = 1$. □

At an endpoint of a fixed-θ segment the complete finite trajectory also has a limit. Indeed Lemma 4.1 gives the initial angle and $p_0 = -z\cos\theta/R$. For each designated next disk, the entry root of the flight quadratic is continuous up to a tangency. The flight length cannot collapse to zero, and the reflection $v \mapsto v - 2(v \cdot n)n$ is continuous at tangency. Induction over the fixed finite itinerary gives the limits of all visited points, velocities, τ_i and c_i. The third-disk clearances have limits as well: the finite horizon restricts the possible nearby disk centres to a finite set, and the minimum distance to each of them is continuous. This argument takes only one-sided limits from the actual event; it asserts no continuation through its boundary.

Theorem 4.3 (Trace completion on the original physical sheets)**.** *On every positive-length original regular pairing segment the coarea factor and weighted density are*

$$\chi = \det D_q(F, \arg g) = e^{\perp\mathsf{T}} G e^{\perp} \ge \lambda_\circ, \qquad f^\varepsilon = \frac{\omega W^\varepsilon}{4\pi c_* \chi}. \tag{4.4}$$

They have finite one-sided endpoint limits. The regular-point variation input (2.3) *implies that f^ε extends absolutely continuously to the closed segment and satisfies*

$$|f^\varepsilon(b-) - f^\varepsilon(a+)| \le C(1 + \varepsilon^{-1}) \int_a^b AI_m/z\, du. \tag{4.5}$$

In fact, on each physical sheet,

$$|f^\varepsilon(v) - f^\varepsilon(u)| \le C(1 + \varepsilon^{-1}) \varrho^{(m-2)_+} |v - u|^{1/2} \quad (a \le u < v \le b). \tag{4.6}$$

If a limiting trajectory at either endpoint contains an actually visited interior grazing collision, its trace at that endpoint is zero. The original paired minimum on that entire segment is then zero, for every original selector.

Proof. The determinant identity in (4.4) follows by expressing $D \arg g$ as $e^{\perp\mathsf{T}} G/z$. The preceding endpoint-limit argument and Lemma 4.2 give limits for G, ω and χ, the last with a positive lower bound. Every factor in the finite product defining W^ε is continuous in its incidence or clearance

argument. Thus f^ε has the asserted finite limits, including at a critical incoming endpoint, where the direction θ is fixed.

Corollary 3.4 and (4.4) bound AI_m by $C\varrho^{(m-2)_+}$. The integral in (4.5) is finite by Lemma 4.1. On compact regular subsegments the flight charts are smooth and the finite clearance minima are locally Lipschitz, so f^ε is locally absolutely continuous. Integrating the inherited regular-point derivative inequality on these subsegments and then passing to the proved endpoint limits establishes absolute continuity and (4.5). The same argument between u and v, using (4.2), gives (4.6).

Finally, if $c_j \to 0$ at a visited interior collision, the marked optical bound gives $\omega \to 0$, while $0 \le W^\varepsilon \le 1$ and $\chi \ge \lambda_\circ$. Hence that endpoint trace is zero. The paired density is its nonnegative minimum with the other trace. □

Remark 4.4 (Two different tangency mechanisms). The last assertion concerns a collision in the visited itinerary. A flight which merely becomes tangent to an *unvisited* third disk has zero clearance but need not have any $c_j = 0$. If all visited incidences remain positive, then (3.7) has a positive limit. A zero third-disk clearance makes $W^\varepsilon = 1$ at the limit; consequently the trace can be positive. Such sheets are not removed. Neither are nongrazing critical entrances or positive section-boundary traces. For a nongrazing critical entrance the exact directional limit is $\omega_* W_*^\varepsilon/(4\pi c_* e^{\perp\mathsf{T}} G_* e^\perp)$. No roof-independent value at the critical point is asserted.

5. The complete physical comparison space

The comparison space is now specified without an unverified endpoint-trace exclusion:

$$\Theta_{\mathrm{ad}} = \Theta_{\mathrm{phys}}. \tag{5.1}$$

More explicitly, it consists of all positive-length maximal regular pairing segments of the original exact event, with their original incident-component identities. Their endpoints may be critical, visited-grazing, third-disk-tangent, or section-boundary limits. Theorem 4.3 supplies finite traces and the endpoint variation inequality on this space. The underlying Borel pairing construction and the averaged analytic estimates remain the inputs in Section 2. No singular trajectory is continued past its physical boundary, and no new source supported at a zero-length segment is introduced.

For fixed $0 < r < \Delta$, define

$$\begin{aligned} q_+(t,\theta) &= \mathbf{1}_{\{0<\ell<\Delta,\ a<t,\ t+r<b\}},\\ q_-(t,\theta) &= (1-q_+)\mathbf{1}_{\{0<\ell<\Delta,\ a<t-r,\ t<b\}}, \qquad Q_r = q_+ + q_-. \end{aligned} \tag{5.2}$$

Thus Q_r is Borel and at most one. Write

$$P_r(t) = \int Q_r p_\vee \, d\theta, \qquad D_r(t) = \int Q_r c^\varepsilon \, d\theta, \qquad E_r(t) = \int Q_r e_\partial \, d\theta.$$

The full pool is retained even on a sheet whose paired source is zero.

Theorem 5.1 (Capacity on the full original comparison space)**.** *At finite collision count, on the original common roof versions,*

$$P_r(t) \le \frac{1}{r}\int_{t-r}^{t+r} \mathsf{T}_\Delta^\varepsilon(u)\,du + C(1+\varepsilon^{-1})\int_{t-\Delta}^{t+\Delta} v^0(u)\,du. \tag{5.3}$$

Consequently, for every previously fixed $K > 0$,

$$\mathcal{H}_M(KP_r) \le CK\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{5.4}$$

Proof. Every positively oriented selected segment contains $[t, t+r]$; every negatively oriented one contains $[t-r, t]$. Integrating its constant paired minimum over the appropriate window gives r times that minimum. Tonelli and $Q_r \le 1$ therefore give $D_r(t) \le r^{-1} \int_{t-r}^{t+r} \mathsf{T}_\Delta^\varepsilon$. Each selected segment lies inside $(t-\Delta, t+\Delta)$. The completed variation inequality (4.5) and the original one-copy physical coarea sum give $E_r(t) \le C(1+\varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0$. Since $P_r = D_r + E_r$, this proves (5.3). Use (2.4) and (2.5), only on these previously fixed windows, to obtain (5.4). The weights are held at the original roof t throughout; none is differentiated or evaluated at the integration roof u. □

5.1. **The physical grazing partition.** For each interior collision let c_i^- and c_i^+ be its incidence limits at the two ends of a pairing segment. Define the Borel sets

$$\Gamma = \bigcup_{i=1}^{m-1} \{c_i^- = 0 \text{ or } c_i^+ = 0\}, \qquad \mathcal{N} = \Theta_{\mathrm{phys}} \setminus \Gamma. \tag{5.5}$$

For $m = 1$ the union is empty. The limits are Borel, for example by evaluating the interior functions at $a + \ell/j$ and $b - \ell/j$ and letting the integer j tend to infinity. Set

$$P_r^{\mathrm{gr}} = \int_\Gamma Q_r p_\vee \, d\theta, \qquad P_r^{\mathrm{ng}} = \int_{\mathcal{N}} Q_r p_\vee \, d\theta, \qquad P_r = P_r^{\mathrm{ng}} + P_r^{\mathrm{gr}}.$$

The superscripts refer to visited collision incidences in the limiting finite itinerary, not to a lower incidence cutoff.

Theorem 5.2 (Zero original grazing source and paid grazing capacity)**.** *For every original Borel selector $0 \le H \le 1$, at finite count,*

$$\int_\Gamma c^\varepsilon \mathbf{1}_{\{a<t<b\}} H(\Psi(\theta,t)) \, d\theta = 0. \tag{5.6}$$

Nevertheless P_r^{gr} need not vanish. It satisfies

$$P_r^{\mathrm{gr}}(t) \le C(1+\varepsilon^{-1}) \int_{t-\Delta}^{t+\Delta} v^0(u) \, du, \tag{5.7}$$

$$\mathcal{H}_M(K P_r^{\mathrm{gr}}) \le CK(1+\varepsilon^{-1})\Delta. \tag{5.8}$$

These estimates include the complete positive capacity on this physical class, with no count of angular blocks or collision words.

Proof. Theorem 4.3 gives $c^\varepsilon = 0$ on Γ, proving (5.6) before the roof pushforward. On that same class $p_\vee = e_\partial$, not necessarily zero. Apply (4.5), use the localization $(a,b) \subset (t-\Delta, t+\Delta)$ on $\{Q_r = 1\}$, and complete the one-copy positive physical sum. This proves (5.7). Equation (2.5) at the fixed width Δ proves (5.8). □

6. The original microscopic shortage after grazing payment

All allocations below are scalar allocations at the original roof; they are not couplings of individual source and comparison sheets. For any original source $U(t)$ define

$$V_{r,K}(t) = \min\{U(t), K P_r(t)\}, \qquad R_{r,K}(t) = (U(t) - K P_r(t))_+.$$

The multiplier is $\beta_{r,K} = \min\{1, KP_r/U\}$ on $U > 0$ and zero on $U = 0$. Then $U = V_{r,K} + R_{r,K}$ at finite count and $V_{r,K}$ is the original source multiplied by $\beta_{r,K}(t)$. When a restricted comparison space is used, source outside it is still in U and may receive this same global fraction. It is not therefore retained sheetwise with coefficient one.

Use the exact source (2.7) and, for a concrete outer choice, put

$$h_B = \delta(B)/2, \qquad r_B = \delta(B)/8, \qquad K_B = \max\{1, \varepsilon(B)^{-1/32}\}. \tag{6.1}$$

For all sufficiently large B these satisfy the inherited cost budget. Define

$$\begin{aligned} \mathsf{R}_B^{\mathrm{full}} &= (\mathsf{U}_B^{\mathrm{micro}} - K_B P_{r_B})_+, \\ \mathsf{R}_B^{\mathrm{ng}} &= (\mathsf{U}_B^{\mathrm{micro}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{6.2}$$

The full capacity pool includes any sheets previously excluded solely for lack of a trace verification. On the complete comparison space it is exactly the canonical pool; on a previously restricted space it is an enlargement, not an identification of different capacities.

Theorem 6.1 (Quantitative payment of the visited-grazing class)**.** *Under the retained physical-source and analytic inputs, for the original event and every fixed M,*

$$0 \le \mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{full}} \le K_B P_{r_B}^{\mathrm{gr}}, \tag{6.3}$$

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{full}}) \le C_M B^{-95/384}, \tag{6.4}$$

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}}) + C_M B^{-1/384}. \tag{6.5}$$

The unchanged pointwise assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}}) \to 0$ for every fixed M. The original microscopic source in these formulas is supported on $\mathcal{N}$ without a change of its positive weight.*

Proof. For $u, a, b \ge 0$, the positive-part function gives $0 \le (u-a)_+ - (u-a-b)_+ \le b$. Apply this with $a = K_B P_{r_B}^{\mathrm{ng}}$ and $b = K_B P_{r_B}^{\mathrm{gr}}$ to prove (6.3) at finite count. Theorem 5.2, $\Delta_B = O(\varepsilon^4)$ and $K_B = O(\varepsilon^{-1/32})$ give

$$\mathcal{H}_M(K_B P_{r_B}^{\mathrm{gr}}) \le C\varepsilon^{-1/32}(1+\varepsilon^{-1})\varepsilon^4 = O(\varepsilon^{95/32}) = O(B^{-95/384}).$$

This proves (6.4) in the prescribed order. By Theorem 5.1, the accepted full-pool source has height at most

$$C\varepsilon^{-1/32}\{\varepsilon^{1/16} + O(\varepsilon^3)\} = O(B^{-1/384}).$$

Insert its exact positive decomposition into (2.8). Since $\mathsf{R}_B^{\mathrm{full}} \le \mathsf{R}_B^{\mathrm{ng}}$, this proves (6.5) and sufficiency of the displayed residual limit. Necessity follows from Input statement 2.2 and $0 \le \mathsf{R}_B^{\mathrm{ng}} \le \mathsf{U}_B^{\mathrm{micro}}$. Equation (5.6) proves the final assertion. □

This theorem supplies a physical component of the required scarcity estimate, rather than a new certificate class. It removes an entire visited-grazing source class exactly and pays the loss of all its comparison capacity by a faster-decaying original optical volume. It does not yet bound $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}})$ by a quantity tending to zero. In $\mathcal{N}$ every visited endpoint incidence is positive at each fixed count, but its infimum may tend to zero with the count. Third-disk tangencies, nongrazing critical entrances, section-boundary cuts, flat and short-reach segments, and the original comparison shortages all remain with their full weight. A collision-uniform estimate on that remaining source is still required for (1.5). Neither finite-count exhaustion nor the per-itinerary exponential bound is used in its place.

7. A positive payment from the variation on a physical segment

The comparison pool of Section 5 is left unchanged. We first make a different use of the completed endpoint estimate: it pays part of the *original source*, whether or not that source has a comparison window. All statements in this section are made on the inherited one-copy physical pairing space.

Write C_{reg} for a fixed constant in (2.3). For fixed $\varepsilon > 0$ define

$$\begin{aligned} \mathcal{V}^{\varepsilon}(\theta) &= \mathrm{Var}_{[a(\theta),b(\theta)]}\, f^{\varepsilon}(\,\cdot\,,\theta), \\ \mathcal{J}(\theta) &= \int_{a(\theta)}^{b(\theta)} \frac{A(u,\theta) I_m(u,\theta)}{z(u,\theta)}\, du. \end{aligned} \tag{7.1}$$

Endpoint values in the variation are the one-sided physical traces, not values assigned on a continuation beyond the original event.

Lemma 7.1 (Measurability and the full variation bound)**.** *Both functions in* (7.1) *are Borel and finite on every positive-length physical segment. Moreover,*

$$\mathcal{V}^{\varepsilon} = \int_a^b |\partial_u f^{\varepsilon}|\, du \le C_{\mathrm{reg}}(1+\varepsilon^{-1})\mathcal{J}. \tag{7.2}$$

This is a bound for total variation, not merely for the difference of the two endpoint values.

Proof. Pull the closed segment back to $[0,1]$ by $u = a + s(b-a)$, using the endpoint traces at $s = 0, 1$. The interior evaluations are jointly Borel. At $s = 0$ the completed value is the limit of evaluations at $a + \ell/j$, $j \ge 2$; at $s = 1$ it is the limit of evaluations at $b - \ell/j$. The completed endpoint values, and hence the function on the closed product, are jointly Borel. The resulting function is continuous in s. Its variation is the supremum of the sums of absolute increments over finite rational partitions containing 0 and 1. Continuity permits this countable family of partitions, so its supremum is Borel. The integral defining $\mathcal{J}$ is Borel by the same change of variables and nonnegative integration. Theorem 4.3 gives absolute continuity on the closed segment. The variation of an absolutely continuous real function is the integral of the absolute value of its derivative. Integrate (2.3); its right side is integrable by Corollary 3.4 and Lemma 4.1. This proves the equality, inequality and finiteness asserted above. □

For $\gamma \ge 1$, split the original paired coefficient into

$$c^{\varepsilon,\gamma}_{\mathrm{var}} = \min\{c^{\varepsilon}, \gamma\mathcal{V}^{\varepsilon}\}, \qquad c^{\varepsilon,\gamma}_{\mathrm{per}} = (c^{\varepsilon} - \gamma\mathcal{V}^{\varepsilon})_+. \tag{7.3}$$

Thus $c^{\varepsilon} = c^{\varepsilon,\gamma}_{\mathrm{var}} + c^{\varepsilon,\gamma}_{\mathrm{per}}$, with both terms nonnegative. The subscript per will be justified by a physical persistence statement in the next section. It denotes a positive *submeasure* of the original source, not a replacement of its coefficient by a different normalization.

For a jointly Borel selector $0 \le H \le 1$ supported on $0 < \ell < h$, set

$$T^{\varepsilon,\gamma}_{H,j}(t) = \int c^{\varepsilon,\gamma}_j(\theta)\mathbf{1}_{\{a<t<b\}}H(\Psi(\theta,t))\, d\theta, \qquad j \in \{\mathrm{var}, \mathrm{per}\}. \tag{7.4}$$

The original selected paired source is their exact sum. The selector is evaluated at t and is not moved to another roof in the proof.

Theorem 7.2 (A summed payment of the positive short source)**.** *At every finite collision count, on the original common roof versions,*

$$0 \le T^{\varepsilon,\gamma}_{H,\mathrm{var}}(t) \le C_{\mathrm{reg}}\gamma(1+\varepsilon^{-1}) \int_{t-h}^{t+h} v^0(u)\, du. \tag{7.5}$$

Consequently, for every $\varepsilon, h > 0$ and $\gamma \ge 1$ fixed before the collision limit,

$$\mathcal{H}_M(T^{\varepsilon,\gamma}_{H,\mathrm{var}}) \le C\gamma(1+\varepsilon^{-1})h. \tag{7.6}$$

The constants are uniform in the collision count, radius, exact labels, segment length and selector. No positive lower bound for ℓ or for an interior incidence is required.

Proof. For a segment meeting t and satisfying $\ell < h$, one has $(a,b) \subset (t-h, t+h)$. By (7.3) and (7.2), its selected contribution is at most

$$C_{\mathrm{reg}}\gamma(1+\varepsilon^{-1})\mathbf{1}_{\{a<t<b,\, \ell<h\}}H(\Psi(\theta,t)) \int_a^b AI_m/z\, du.$$

First sum these nonnegative quantities over the physical sheets. Tonelli's theorem applies, without an estimate for the number of sheets. At an integration roof u, each retained pair (θ, u) is still a

point of the same original exact event. Dropping its selector, which is between zero and one, and using physical multiplicity one therefore bounds the sum by

$$C_{\mathrm{reg}}\gamma(1+\varepsilon^{-1})\int_{t-h}^{t+h}\int_{\{a(\theta)<u<b(\theta)\}} A(u,\theta)I_m(u,\theta)/z(u,\theta)\,d\theta\,du.$$

The coarea identity in Input statement 2.1 identifies the inner integral with $v^0(u)$. This proves the finite-count inequality. Apply (2.5) at the previously fixed width h, using $\int_{t-h}^{t+h} v^0 = 2hA_h v^0$. Only after the complete sum and this inequality are established are the uniform norms and collision limsup taken. This proves (7.6). □

Corollary 7.3 (The original microscopic schedule)**.** *Keep $\widetilde{H}_B, h_B, r_B, K_B$ exactly as in* (2.7) *and* (6.1)*, and choose*

$$\gamma_B = \max\{2, \varepsilon(B)^{-1/32}\}. \tag{7.7}$$

Let $\mathsf{U}_B^{\mathrm{var}}$ and $\mathsf{U}_B^{\mathrm{per}}$ denote the two sources in (7.4) *with $H = \widetilde{H}_B$, $\varepsilon = \varepsilon(B)$ and $\gamma = \gamma_B$. Then*

$$\mathsf{U}_B^{\mathrm{micro}} = \mathsf{U}_B^{\mathrm{var}} + \mathsf{U}_B^{\mathrm{per}}, \qquad \mathcal{H}_M(\mathsf{U}_B^{\mathrm{var}}) \le C_M B^{-95/384}. \tag{7.8}$$

Proof. The identity holds before pushforward, by (7.3). For sufficiently large B, $h_B = O(\varepsilon^4)$ and $\gamma_B = O(\varepsilon^{-1/32})$. Theorem 7.2 gives the bound

$$C\gamma_B(1+\varepsilon^{-1})h_B = O(\varepsilon^{95/32}) = O(B^{-95/384}).$$

All three quantities $h_B, \varepsilon(B), \gamma_B$ are fixed before m tends to infinity. No collision-dependent residence is inserted into an estimate asserted only for fixed residence. □

Unlike (5.6), this payment does not rely on a zero paired coefficient. If $c^\varepsilon > 0$ and $\mathcal{V}^\varepsilon > 0$, its coefficient in (7.3) is positive. If $c^\varepsilon \le \gamma\mathcal{V}^\varepsilon$, the complete original coefficient is paid. These statements do not assert a positive uniform proportion of paid mass in every exact event.

8. Persistent guarded segments and the same pointwise target

The positive decomposition of the preceding section has a geometric consequence on every segment which retains source. It uses the original incidence and clearance guard, without adding a distance to the section boundary.

Theorem 8.1 (Persistence of the guard on the remaining source)**.** *Let $\gamma > 1$. If $c^{\varepsilon,\gamma}_{\mathrm{per}}(\theta) > 0$, then $c^\varepsilon > 0$ and, at every point of the closed physical segment,*

$$|f^\varepsilon(u,\theta) - c^\varepsilon(\theta)| \le \mathcal{V}^\varepsilon(\theta) < \gamma^{-1}c^\varepsilon(\theta). \tag{8.1}$$

In particular $f^\varepsilon > 0$ and $W^\varepsilon > 0$ throughout that segment, including its one-sided endpoints. For each u at least one of the original tests satisfies

$$c_j(u) < 2\varepsilon q_0^{d_j/4} \quad \text{for some } 0 \le j \le m, \qquad \text{or} \qquad \Delta_j(u) < 2\varepsilon q_0^{d'_j/4} \quad \text{for some } 0 \le j < m. \tag{8.2}$$

The witnessing test may change with u.

Conversely, if W^ε vanishes anywhere on the closed segment, its complete original paired coefficient is paid: $c^{\varepsilon,\gamma}_{\mathrm{var}} = c^\varepsilon$ and $c^{\varepsilon,\gamma}_{\mathrm{per}} = 0$.

Proof. The endpoint minimum is attained at at least one endpoint, where $f^\varepsilon = c^\varepsilon$. Every increment from that endpoint is bounded by the total variation. Positivity of the remaining coefficient is precisely $c^\varepsilon > \gamma\mathcal{V}^\varepsilon$, which proves (8.1). If any visited interior incidence vanished at an endpoint, Theorem 4.3 would give $c^\varepsilon = 0$. Thus the segment belongs to $\mathcal{N}$. Its optical factor A is finite and positive on the closed segment: the exact formula $\omega = Rc_0c_m/\mathfrak{b}_m$ in (3.7) has a finite positive denominator when all visited incidences are positive, while (4.4) gives $\chi \ge \lambda_\circ$. This also applies at a critical endpoint with its fixed Gauss direction. Since $f^\varepsilon = AW^\varepsilon > 0$, the

guard is positive there. If all inequalities in (8.2) failed, every factor defining H^ε would equal one, contradicting $W^\varepsilon = 1 - H^\varepsilon > 0$.

For the converse, let u_0 be a point where $W^\varepsilon = 0$. The finite optical factor gives $f^\varepsilon(u_0) = 0$, so the increment from an endpoint attaining c^ε gives $\mathcal{V}^\varepsilon \geq c^\varepsilon$. Apply (7.3). This argument also covers $u_0 = a$ or $u_0 = b$, with the proved physical one-sided values. □

The surviving class

$$\mathcal{P}_B = \{\theta \in \mathcal{N} : c^{\varepsilon(B)}(\theta) > \gamma_B \mathcal{V}^{\varepsilon(B)}(\theta)\} \tag{8.3}$$

is Borel. The source $\mathsf{U}_B^{\mathrm{per}}$ is supported on it; its coefficient is still the positive difference in (7.3), not the full original coefficient. The sum (7.8) retains every original coefficient exactly. No assumption that one fixed bad test covers the entire segment is made. A Borel witness at a specified point can be chosen as the first successful test in the finite ordered list in (8.2); this selection is not needed to prove the payment.

Remark 8.2 (Boundary mechanisms and what is retained). An unvisited third-disk tangency may give $W^\varepsilon = 1$ at an endpoint without forcing variation relative to the endpoint minimum. It is therefore not declared paid merely on account of its name. The same applies to a nongrazing critical entrance, an original section cut and a short-reach segment. Each receives precisely the two coefficients in (7.3). When the guard vanishes elsewhere on that segment the entire coefficient is paid by Theorem 8.1; otherwise a positive remaining coefficient is possible. At fixed collision count the optical data on a surviving segment have positive minima, but no such minimum is asserted uniformly in the collision count or the sheet.

8.1. **Comparison with the original nongrazing shortage.** Keep the *full original* nongrazing pool $P_{r_B}^{\mathrm{ng}}$, including its capacity on segments whose source has already been paid. Define only a new source-side remainder:

$$\mathsf{R}_B^{\mathrm{per}}(t) = \big(\mathsf{U}_B^{\mathrm{per}}(t) - K_B P_{r_B}^{\mathrm{ng}}(t)\big)_+. \tag{8.4}$$

The allocation by this pool is scalar at the roof, just as in Section 6. It is not a coupling of individual persistent sheets to individual comparison sheets.

Theorem 8.3 (Positive-source reduction with a paid error)**.** *On the original exact event, at every finite collision count,*

$$\mathsf{U}_B^{\mathrm{micro}} = \mathsf{U}_B^{\mathrm{var}} + \min\{\mathsf{U}_B^{\mathrm{per}}, K_B P_{r_B}^{\mathrm{ng}}\} + \mathsf{R}_B^{\mathrm{per}}, \tag{8.5}$$

$$0 \leq \mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{per}} \leq \mathsf{U}_B^{\mathrm{var}}. \tag{8.6}$$

For every fixed M, in the prescribed order of limits,

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{per}}) \leq C_M B^{-95/384}, \tag{8.7}$$

$$\mathcal{E}_M \leq \mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}}) + C_M B^{-1/384}. \tag{8.8}$$

The unchanged pointwise assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}}) \to 0$ for every fixed M.*

Proof. Equation (8.5) is the positive scalar identity $x = \min(x, y) + (x - y)_+$ applied to $x = \mathsf{U}_B^{\mathrm{per}}$, combined with (7.8). Monotonicity and the unit Lipschitz bound of the positive-part function give (8.6). Corollary 7.3 then proves (8.7).

For completeness, the height of the middle term in (8.5) is at most

$$\mathcal{H}_M(K_B P_{r_B}) \leq C\varepsilon^{-1/32}\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})\Delta_B\} = O(\varepsilon^{1/32}) = O(B^{-1/384}),$$

by Theorem 5.1. Insert the three nonnegative terms in Input statement 2.2, and use subadditivity of the ordered upper height. This proves (8.8) and sufficiency of the displayed limit. Necessity follows from the same inherited scalar equivalence and $0 \leq \mathsf{R}_B^{\mathrm{per}} \leq \mathsf{U}_B^{\mathrm{micro}}$. No essential supremum is interchanged with an integral, and the comparison pool has not been restricted to $\mathcal{P}_B$. □

The estimate still required to conclude (1.5) is

$$\lim_{B\to\infty}\limsup_{m\to\infty}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \left(\mathsf{U}^{\mathrm{per}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\right)_+ = 0. \tag{8.9}$$

Theorems 7.2 and 8.1 prove an actual summed payment and a necessary physical persistence property. They do not prove (8.9). In particular a positive nearly constant density on a shrinking segment need not have small pointwise height merely because its roof integral is small. The original raw pointwise theorem, and not an averaged substitute, remains the organizing target.

9. Global uniqueness on a roof–angle fiber

The estimates in this section concern actual finite trajectories, not a continuation through a forbidden event. In particular the regular domain of a lifted itinerary need not be connected or convex. This distinction is important when the original section and occupation tests divide that domain into many components.

Normalize the initial lifted scatterer to $O_0 = 0$. Let $\mathcal{W}_m(R)$ denote the realizable lifted itineraries of m flights. This set is finite by the finite-horizon bound and the lattice geometry. Fix $w = (O_0, \dots, O_m) \in \mathcal{W}_m(R)$. For a physical trajectory write $Q_j = O_j + Rn_j$, let t_j be the counterclockwise rotation of n_j, and put

$$S(Q) = \sum_{j=0}^{m-1} |Q_{j+1} - Q_j|, \qquad g = R(-p_0, p_m), \qquad z = |g|.$$

As a function of the independent Cartesian contacts, S is convex. It is differentiable at every configuration in question because successive scatterer disks are disjoint. Write $\beta_0 = \beta_m = 1$ and $\beta_j = 2$ for $0 < j < m$.

Lemma 9.1 (Two-trajectory support inequalities)**.** *Let Q and $\widehat{Q}$ be two regular physical trajectories with the same lifted itinerary and radius. Set*

$$b_j = 1 - n_j \cdot \widehat{n}_j \geq 0, \qquad s_j = t_j \cdot \widehat{n}_j, \qquad s_E = (s_0, s_m).$$

Their lengths satisfy

$$S(\widehat{Q}) - S(Q) \geq R\sum_{j=0}^{m} \beta_j c_j b_j + g \cdot s_E, \tag{9.1}$$

$$S(Q) - S(\widehat{Q}) \geq R\sum_{j=0}^{m} \beta_j \widehat{c}_j b_j - \widehat{g} \cdot s_E. \tag{9.2}$$

Consequently

$$(\widehat{g} - g) \cdot s_E \geq R\sum_{j=0}^{m} \beta_j (c_j + \widehat{c}_j) b_j. \tag{9.3}$$

No interpolation by physical trajectories is used.

Proof. The physical reflection law and first variation in Cartesian coordinates give

$$\nabla_0 S = -c_0 n_0 - p_0 t_0, \qquad \nabla_j S = -2c_j n_j \ (0 < j < m), \qquad \nabla_m S = -c_m n_m + p_m t_m.$$

The convex support inequality is $S(\widehat{Q}) - S(Q) \geq \sum_j \nabla_j S(Q) \cdot (\widehat{Q}_j - Q_j)$. Since $\widehat{Q}_j - Q_j = R(\widehat{n}_j - n_j)$, its normal terms are $R\beta_j c_j b_j$ and its two tangential terms are $g \cdot s_E$. This proves (9.1). Reversing the roles of the two configurations leaves b_j unchanged and replaces s_j by $-s_j$. This proves (9.2). Adding the two inequalities proves (9.3). The support inequality is applied to a convex Cartesian function; it requires neither a convex angular chart nor a physical path joining the configurations. □

Theorem 9.2 (Global roof–angle uniqueness)**.** *For a fixed lifted itinerary, radius and direction $e \in S^1$, the physical length is strictly increasing among regular trajectories whose endpoint gradients are $g = ze$, $z > 0$, ordered by z. At a fixed z there is at most one such trajectory. In particular*

$$Q \longmapsto (F(Q), \arg g(Q)) \tag{9.4}$$

is globally injective on the noncritical regular trajectories of that itinerary. This remains true after any of the original exact event tests, and across all their disconnected regular components.

Proof. If $\widehat{g} = g$, the left side of (9.3) is zero. All c_j and $\widehat{c}_j$ are strictly positive, so every $b_j = 0$. Thus all physical contacts, and then all flight directions, coincide.

Now let $\widehat{g} = \widehat{z}e$, $g = ze$ and $\widehat{z} > z > 0$. The configurations cannot coincide. At least one b_j is therefore positive, and (9.3) gives

$$(\widehat{z} - z)e \cdot s_E \geq R \sum_j \beta_j (c_j + \widehat{c}_j) b_j > 0.$$

Hence $e \cdot s_E > 0$. The right side of (9.1) is then strictly positive, proving $S(\widehat{Q}) > S(Q)$. Equal roof and angle would force equal z, and the first part of the proof would force equality of trajectories. Restricting an injective map to the original exact event cannot increase its multiplicity. □

Lemma 9.3 (A bounded roof range for each physical itinerary)**.** *For any two regular physical trajectories with the same itinerary,*

$$|S(Q) - S(\widehat{Q})| \leq |Q_0 - \widehat{Q}_0| + |Q_m - \widehat{Q}_m| \leq 4R. \tag{9.5}$$

Proof. With its endpoint contacts fixed, a regular physical trajectory minimizes S over the product of the closed interior scatterer disks. Indeed, for a competitor Y_j in the jth disk, $n_j \cdot (Y_j - Q_j) \leq 0$; the convex support inequality and $\nabla_j S = -2c_j n_j$ therefore give the minimizing property. Use the interior contacts of $\widehat{Q}$ as a competitor for the endpoints of Q. Changing only the first and last endpoint changes the sum of flight lengths by at most the sum of their displacements, by the triangle inequality. This proves one direction of (9.5); interchanging the trajectories proves the other. The same triangle inequality covers $m = 1$. □

This range estimate does not control how many original occupation labels occur in a word domain. In particular it is not used to replace the exact occupation test by an unqualified interval test.

9.1. **The original physical sum at a target roof.** For the original section tests, $\chi \geq \lambda_\circ$ by (4.4). On each regular component, the map (9.4) is a local diffeomorphism. Theorem 9.2 shows that all those components of a fixed itinerary have disjoint images, not merely that each component has a nonsingular Jacobian. The equal-gradient argument also gives at most one regular critical trajectory per word, even when the gradient is zero. There are finitely many physical words at each finite count, so these critical points have zero collision area. Consequently, for any nonnegative Borel trajectory weight a, the original roof density has the representation

$$\frac{dF_\#(a\mathbf{1}_\Pi \nu_R^*)}{dt}(t) = \int_0^{2\pi} \sum_{w \in \mathcal{W}_m(R)} \mathbf{1}_{\{x_w(t,\vartheta) \text{ exists}\}} \frac{\omega(x_w) a(x_w)}{4\pi c_* \chi(x_w)} \, d\vartheta. \tag{9.6}$$

Here x_w is the unique point of the original exact event with the indicated word, roof and angle. A missing point contributes zero. The representation holds for almost every roof, in the original order $\sup_R \operatorname{ess\,sup}_t$. Angular chart overlap is removed before summation; the incident-component convention remains unchanged.

Proposition 9.4 (Monotone completion of the target-roof formula)**.** *Formula* (9.6) *is an identity of nonnegative measures for every nonnegative Borel weight, with extended values allowed before an integrability bound is proved. In particular it applies to the unbounded products* $\prod_{i\in S} c_i^{-1}$.

Proof. Fix m, R, n, k. Cover the noncritical regular physical domain in endpoint coordinates by countably many relatively compact local charts V_j on which $(F, \arg g)$ is a C^1 diffeomorphism into an unwrapped angular chart. Such a cover exists by the nonvanishing Jacobian and second countability. Subdivide as necessary so that both maps have bounded derivatives on each smaller chart. Disjointize the source charts by $B_j = V_j \setminus \bigcup_{k<j} V_k$. The sets B_j are Borel; no chart overlap is counted twice. Angular representatives are also split at a fixed seam, so their range is $[0, 2\pi)$.

For a nonnegative Borel weight h put

$$h_N = \min\{h, N\}\mathbf{1}_{\bigcup_{j\le N} B_j}.$$

This sequence increases to h on the regular noncritical domain. Apply ordinary change of variables on each of the first N charts, restricted to its Borel piece, and test against a nonnegative Borel function of (t, ϑ). The local inverse Jacobian, multiplied by the original collision density, is exactly $\omega/(4\pi c_*\chi)$. Local inverses are continuous and agree where they represent the same physical point, so their Borel restrictions define the displayed x_w without a selection ambiguity. Global uniqueness prevents different regular components of one word from producing the same target. Distinct words are summed, not identified.

Monotone convergence on the source, followed by Tonelli and monotone convergence on the target, now gives the identity for h as an extended nonnegative measure. Only after this passage is an upper pressure bound used to establish finiteness and an L^∞ density. This avoids assuming the integrability of the unbounded marked weight in order to prove its own change of variables.

The complement of the two-sided regular set has zero collision area. On the countably many charts just chosen, absolute continuity of the coordinate changes and their local Lipschitz property give zero target area for its image. The finitely many regular critical points also have zero source area. Removing these sets therefore does not change the weighted source measure. Fubini yields the claimed roof identity for almost every t. All null sets here are chosen at the fixed radius; no exceptional set common to every R is asserted or needed. For the finitely many marked sets at a fixed m, intersecting the corresponding full-measure roof sets provides a common version for their finite positive expansions. □

10. A complete marked pressure bound at the target roof

We now transfer the retained geometric pressure estimate to the roof fibers in (9.6). The previous momentum-fiber bound alone would not justify this transfer: different words at the same roof can have different endpoint momentum magnitudes. The global injectivity just proved supplies the missing multiplicity statement.

Let $j(x) = J^s T_R(x)$ on the full-area set M'_R regular in both infinite time directions, and let $j_l(x) = \prod_{v=0}^{l-1} j(T_R^v x)$, with $j_0 = 1$. The lifted physical one-flight alphabet has at most D members, uniformly in R. Define

$$P_l(R) = \sum_{w\in\mathcal{W}_l(R)} \sup_{x\in[w]_R\cap M'_R} j_l(x), \qquad P_0 = 1.$$

Input statement 10.1 (Uniform physical pressure and stable optical comparison)**.** On the circular family $R \in [9/20, 47/100]$, use the original finite-horizon flight bounds, positive curvature bounds and the original endpoint section cap. The stable Jacobian j is computed with Euclidean arclength along the true stable line in collision coordinates (s, φ), where $s = R\alpha$ and $p = \sin\varphi$. It is restricted to M'_R, the full-area set regular in both infinite directions. Its cocycle uses this

same convention on every block, including the empty block. The endpoint weight ω is still the derivative in the original (α, p) coordinates of (2.1); all coordinate factors are included in the uniform optical constant.

For the realizable lifted physical words just defined, the retained geometric theorems state

$$\sup_{R,l} P_l(R) \le Q_*, \qquad \omega(x) \le C_0 c_m(x) j_m(x), \qquad j(x)/c(x) \le C_1. \tag{10.1}$$

All three constants are independent of R, l and the original exact labels. These statements concern positive, unconditioned physical word sums; they are not a twisted or exact-label pressure theorem.

We take $Q_* \ge 1$ and $C_0, C_1 > 0$. These are the uniform geometric pressure theorem in historical module 241 and the stable optical comparison in module 238, with the physical-word domination in module 239. Their complete original sources are pinned in the proof ledger and copied in the provenance directory. They involve no occupation, displacement or roof test. Such tests are dropped only on the upper side of the estimates below. No new pressure estimate for a twisted or point-conditioned operator is assumed.

Lemma 10.2 (Removal of several marked stable factors)**.** *For $S = \{i_1 < \cdots < i_s\} \subset \{1, \dots, m-1\}$, at every regular trajectory in M_R',*

$$\frac{\omega(x)}{\prod_{i\in S} c_i(x)} \le C_0 C_1^s \prod_{v\in\{0,\dots,m-1\}\setminus S} j(T_R^v x). \tag{10.2}$$

Consequently, if at most one point x_w is chosen from each physical word domain, then

$$\sum_w \frac{\omega(x_w)}{\prod_{i\in S} c_i(x_w)} \le C_0 C_1^s D^s Q_*^{s+1}. \tag{10.3}$$

The choice of point may depend on the word and need not have the same momentum target for different words.

Proof. Use $c_m \le 1$ in (10.1), factor j_m into its individual steps, and apply $j(T_R^i x)/c_i(x) \le C_1$ at each marked index. This gives (10.2).

Delete the s marked steps of a word. The remaining blocks have lengths

$$i_1, \quad i_2 - i_1 - 1, \ \dots, \ i_s - i_{s-1} - 1, \quad m - i_s - 1.$$

For clarity, steps are numbered $0, \dots, m-1$, whereas the marks belong to $1, \dots, m-1$. Their indexing is

$$\underbrace{0, \dots, i_1 - 1}_{i_1} \mid i_1 \mid \underbrace{i_1 + 1, \dots, i_2 - 1}_{i_2 - i_1 - 1} \mid i_2 \mid \cdots \mid i_s \mid \underbrace{i_s + 1, \dots, m-1}_{m - i_s - 1}.$$

For example, $m = 7$ and $S = \{1, 2, 6\}$ leave blocks of lengths $(1, 0, 3, 0)$ and delete three physical symbols. Thus adjacent marks and a final mark are included, and every empty block contributes $P_0 = 1$. Some lengths can be zero. The blocks, translated to the initial fundamental scatterer, together with the s deleted physical step symbols determine the full lifted word uniquely. There are at most D^s choices for those symbols. Bound the remaining cocycle on each block by its own supremum and sum over all block words. Positivity permits adding combinations which do not produce a realizable full word. The resulting upper bound is $C_0 C_1^s D^s$ times the product of the $s+1$ corresponding $P_l(R)$'s, at most $C_0 C_1^s D^s Q_*^{s+1}$. For $S = \varnothing$ it is simply $C_0 P_m(R)$. This argument counts physical step symbols, never the infinitely many homogeneous strips at a removed incidence. □

Set

$$C_\star = \frac{C_0 Q_*}{2c_* \lambda_\circ}, \qquad K_\star = C_1 D Q_*. \tag{10.4}$$

Neither constant depends on the collision count, radius, exact labels, roof or marked indices.

Theorem 10.3 (Target-roof multiple-incidence insertion)**.** *For every finite m, every original exact event and every $S \subset \{1, \dots, m-1\}$, the nonnegative measure*

$$F_\# \left(\prod_{i \in S} c_i^{-1} \mathbf{1}_\Pi \nu_R^* \right)$$

has a roof density d_S satisfying

$$\|d_S\|_{L^\infty(dt)} \le C_\star K_\star^{|S|}. \tag{10.5}$$

The bound holds on the whole roof line. For arbitrary numbers $a_1, \dots, a_{m-1} \ge 0$, the density $d_{\boldsymbol{a}}$ with weight $\prod_{i=1}^{m-1}(1 + a_i/c_i)$ satisfies

$$\|d_{\boldsymbol{a}}\|_\infty \le C_\star \prod_{i=1}^{m-1} (1 + K_\star a_i) \le C_\star \exp\left(K_\star \sum_{i=1}^{m-1} a_i \right). \tag{10.6}$$

In particular the complete inherited linear incidence weight obeys

$$\left\| \frac{dF_\#(I_m \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \le C_\star \left(1 + K_\star \frac{2q_0}{1 - q_0} \right). \tag{10.7}$$

These are finite-count unscaled height bounds, not m^{-2} local limit estimates.

Proof. At an almost-everywhere target (t, ϑ), Theorem 9.2 supplies at most one x_w per word. Apply (10.3) to precisely those points, use $\chi \ge \lambda_\circ$, and integrate (9.6) over the angle interval of length 2π. This gives (10.5) with the constants (10.4). Proposition 9.4 first establishes the nonnegative change-of-variables identity by truncation and monotone exhaustion; the present bound then proves finiteness and the density. At finite m the roof is confined to a bounded interval.

Expand $\prod_i (1 + a_i/c_i)$ into its finite sum over distinct marked sets. Apply (10.5) to each term. The sum of the upper bounds factors exactly as $C_\star \prod_i (1 + K_\star a_i)$. The exponential inequality uses $1 + x \le e^x$. Finally expand I_m linearly and use $\sum_i q_0^{d_i} \le 2q_0/(1 - q_0)$. All original exact labels remain in the source throughout. Removing their tests only enlarges the positive word upper sums in the proof; it does not change the source normalization or infer a Gaussian factor from an unconditioned pressure bound. □

Corollary 10.4 (A factorial bound for changing incidence witnesses)**.** *Let $J \subset \{1, \dots, m-1\}$ and let $\eta_i \ge 0$ be arbitrary. On the original event define*

$$N_J(x) = \sum_{i \in J} \mathbf{1}_{\{c_i(x) < \eta_i\}}, \qquad A_J = \sum_{i \in J} \eta_i.$$

For every integer $s \ge 1$,

$$\left\| \frac{dF_\#(\mathbf{1}_{\{N_J \ge s\}} \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \le C_\star K_\star^s e_s((\eta_i)_{i \in J}) \le C_\star \frac{(K_\star A_J)^s}{s!}, \tag{10.8}$$

where $e_s = 0$ for $s > |J|$. The witnesses may change with the target roof and angle; no single witness is chosen for an entire segment.

Proof. At every regular state,

$$\mathbf{1}_{\{N_J\ge s\}} \le \binom{N_J}{s} = \sum_{\substack{S\subset J\\|S|=s}} \prod_{i\in S} \mathbf{1}_{\{c_i<\eta_i\}} \le \sum_{\substack{S\subset J\\|S|=s}} \prod_{i\in S} \eta_i/c_i.$$

Push this positive inequality through the original roof and apply (10.5). In the expansion of A_J^s each product with distinct indices occurs $s!$ times; all remaining terms are nonnegative. This proves the last inequality. The argument is finite-count and already completes the full physical sum, so it introduces no collision-count multiplier for the possible witnesses. □

11. Pointwise removal of deep and numerous incidence witnesses

We apply the marked pressure theorem to the unchanged positive persistent source of Section 8. The new payment concerns its value at the original target roof, rather than an average in that roof. Clearance witnesses require a separate estimate and will not be identified with incidence witnesses.

Lemma 11.1 (Domination of the surviving coefficient at every roof)**.** *For $\gamma \ge 1$, on every original physical segment and every point of its closed interval,*

$$0 \le c_{\mathrm{per}}^{\varepsilon,\gamma} \le f^{\varepsilon}(u,\theta) \le A(u,\theta). \tag{11.1}$$

Proof. Total variation bounds the decrement from an endpoint attaining c^{ε}, so $f^{\varepsilon}(u) \ge c^{\varepsilon} - \mathcal{V}^{\varepsilon}$. If $c_{\mathrm{per}} > 0$, its value is $c^{\varepsilon} - \gamma\mathcal{V}^{\varepsilon} \le c^{\varepsilon} - \mathcal{V}^{\varepsilon}$. If $c_{\mathrm{per}} = 0$, nonnegativity suffices. The last inequality is the unchanged bound $0 \le W^{\varepsilon} \le 1$. □

Thus a target-dependent Borel restriction of the persistent paired source is bounded by the corresponding restriction of the original uncut roof measure. This assertion concerns the surviving coefficient, not the full endpoint minimum on an arbitrary segment.

For $\varepsilon > 0$, use precisely the incidence witness thresholds already forced by the original guard:

$$E_i^{\varepsilon} = \{c_i < \eta_i^{\varepsilon}\}, \qquad \eta_i^{\varepsilon} = 2\varepsilon q_0^{d_i/4}, \quad d_i = \min(i, m-i), \quad 1 \le i < m. \tag{11.2}$$

Write $N^{\varepsilon} = \sum_{i=1}^{m-1} \mathbf{1}_{E_i^{\varepsilon}}$. There are at most two indices at each depth. Hence, for an integer $L \ge 1$,

$$\sum_{d_i\ge L} \eta_i^{\varepsilon} \le \frac{4\varepsilon q_0^{L/4}}{1-q_0^{1/4}}, \qquad \sum_{i=1}^{m-1} \eta_i^{\varepsilon} \le A_{\varepsilon} := \frac{4\varepsilon q_0^{1/4}}{1-q_0^{1/4}}. \tag{11.3}$$

For any Borel target event E, let

$$\mathsf{U}_B^{\mathrm{per}}[E](t) = \int c_{\mathrm{per}}^{\varepsilon(B),\gamma_B}(\theta)\mathbf{1}_{\{a<t<b\}}\widetilde{H}_B(\Psi(\theta,t))\mathbf{1}_E(\Psi(\theta,t))\,d\theta.$$

These are restrictions of the source in (7.8); no new normalization is used.

Theorem 11.2 (Uniform target-roof witness estimates)**.** *For every finite m, every B and integers $L, s \ge 1$,*

$$\|\mathsf{U}_B^{\mathrm{per}}[\bigcup_{d_i\ge L} E_i^{\varepsilon(B)}]\|_{\infty} \le \frac{4C_\star K_\star \varepsilon(B)}{1-q_0^{1/4}} q_0^{L/4}, \tag{11.4}$$

$$\|\mathsf{U}_B^{\mathrm{per}}[N^{\varepsilon(B)} \ge s]\|_{\infty} \le C_\star \frac{(K_\star A_{\varepsilon(B)})^s}{s!}. \tag{11.5}$$

The norms are essential suprema on the whole roof line; constants are uniform in the radius and original exact labels.

Fix any $\eta > 0$ and define, at finite count,

$$\begin{aligned} L_m &= \left\lceil \frac{4(2+\eta)}{|\log q_0|} \log(m+1) \right\rceil, \\ s_m &= \left\lceil \frac{4\log(m+e)}{\log\log(m+e^e)} \right\rceil, \\ \mathcal{D}_{B,m} &= \bigcup_{d_i \geq L_m} E_i^{\varepsilon(B)}, \qquad \mathcal{A}_{B,m} = \{N^{\varepsilon(B)} \geq s_m\}, \qquad \mathcal{Z}_{B,m} = \mathcal{D}_{B,m} \cup \mathcal{A}_{B,m}. \end{aligned} \tag{11.6}$$

An empty union contributes zero. For every fixed outer B and M,

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}]) = 0. \tag{11.7}$$

More precisely the deep part satisfies, at every count,

$$\sup_{R,n,k} m^2 \|\mathsf{U}_B^{\mathrm{per}}[\mathcal{D}_{B,m}]\|_\infty \leq C\varepsilon(B)(m+1)^{-\eta}. \tag{11.8}$$

Proof. By Lemma 11.1, each source in the first two assertions is bounded by the original uncut density restricted to the same target event. Apply Corollary 10.4 with $s = 1$ to the depth tail and with general s to all indices, then use (11.3). This proves the two finite-count bounds after the physical sum, with no selected uniform witness.

They hold for every L and s at every m. Substituting L_m in the first gives $q_0^{L_m/4} \leq (m+1)^{-(2+\eta)}$, which proves (11.8). For fixed B, put $a = K_\star A_{\varepsilon(B)} < \infty$. The elementary factorial bound $s! \geq (s/e)^s$ gives

$$m^2 \frac{a^{s_m}}{s_m!} \leq \exp\{2\log m + s_m(1 + \log a - \log s_m)\} = m^{-2+o(1)} \longrightarrow 0.$$

Here $s_m \log s_m = (4+o(1))\log m$ and $s_m = o(\log m)$. The case $a = 0$ is immediate. If $s_m > m-1$, the event is empty. Subadditivity and positivity complete the proof of (11.7).

The depth and multiplicity thresholds in this proof are substitutions into finite-count inequalities valid simultaneously for all their values. No fixed-window spectral estimate is evaluated at a collision-dependent window. The original outer quantities $\varepsilon(B), h_B, r_B, \gamma_B, K_B$ remain fixed before the collision limit. Thus the proof does not exchange that limit with any outer width or band. □

11.1. **The retained source after this pointwise payment.** Define

$$\begin{aligned} \mathsf{U}_B^{\mathrm{loc}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c], \\ \mathsf{U}_B^{\mathrm{exc}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}], \qquad \mathsf{R}_B^{\mathrm{loc}} = (\mathsf{U}_B^{\mathrm{loc}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{11.9}$$

The superscript loc records localization of all incidence witnesses to the two collision-end layers; it makes no claim that clearance witnesses are localized to those layers. The comparison pool is the same full nongrazing pool, without a source-dependent restriction.

Corollary 11.3 (Exact preservation and collision-zero residual difference)**.** *At finite count the original persistent source satisfies*

$$\mathsf{U}_B^{\mathrm{per}} = \mathsf{U}_B^{\mathrm{exc}} + \mathsf{U}_B^{\mathrm{loc}}, \qquad 0 \leq \mathsf{R}_B^{\mathrm{per}} - \mathsf{R}_B^{\mathrm{loc}} \leq \mathsf{U}_B^{\mathrm{exc}}. \tag{11.10}$$

For every fixed B, M,

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}} - \mathsf{R}_B^{\mathrm{loc}}) = 0, \qquad \mathcal{E}_M \leq \mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}}) + C_M B^{-1/384}. \tag{11.11}$$

The unchanged raw pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}}) \to 0$ for every fixed M.

Proof. The source partition is an identity of nonnegative original submeasures. Monotonicity and the unit Lipschitz bound of the positive-part map give its residual inequality. Apply (11.7) and then Theorem 8.3. For necessity use $0 \le \mathsf{R}_B^{\mathrm{loc}} \le \mathsf{U}_B^{\mathrm{micro}}$ and the retained scalar equivalence. Every removed term has been retained in the identity and paid in the target essential-height topology. □

For all sufficiently large B, $2\varepsilon(B) < c_\circ$, so the original endpoint section cap rules out incidence witnesses at $i = 0, m$. At a target point with positive surviving coefficient the original guard still requires an incidence or a clearance witness, by Theorem 8.1. On the retained part, every incidence witness has depth below L_m, and fewer than s_m such witnesses occur. A clearance witness may occur at any original flight, with its original positive weight. These alternatives allow a witness which changes with the roof; they assert no common witness on an entire physical segment.

The precise remaining estimate is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}_{B,m}^{\mathrm{loc}}(t) - K_B P_{r_B,m}^{\mathrm{ng}}(t))_+ = 0. \tag{11.12}$$

The target-roof pressure theorem proves a new physical weighted summation and the pointwise removal in Theorem 11.2; it does not prove (11.12). In particular the all-incidence bound $C\varepsilon$ furnished by (11.4) with $L = 1$ is unscaled. At a fixed outer B it cannot be multiplied by m^2 and declared small. Nor does an inverse-incidence insertion estimate an inverse third-disk clearance. The remaining source includes the original shallow incidence witnesses, unvisited tangencies, critical entrances, section cuts and capacity deficiencies which are not paid by the proved estimates. No change of the original theorem or of its arithmetic classes is made in stating this boundary.

12. Joint depth and severity of incidence witnesses

The factorial estimate treats the witness thresholds only through their sum. It therefore forgets that there are at most two collision indices at any depth. It also forgets how much an individual incidence lies below its guard threshold. We retain both pieces of information. All estimates below concern the same target-roof measure as Theorem 10.3, with its original exact labels. They use its inherited geometric inputs, not a new conditional mixing or local-limit assumption.

Put

$$r = q_0^{1/4}, \qquad \alpha = -\log r = \tfrac{1}{4}|\log q_0|, \qquad b_\varepsilon = 4K_\star\varepsilon/\alpha. \tag{12.1}$$

For a regular trajectory define its incidence-witness set and load by

$$\begin{aligned} J_\varepsilon(x) &= \{i : 1 \le i < m,\ c_i(x) < 2\varepsilon r^{d_i}\}, \\ \mathscr{L}_\varepsilon(x) &= \sum_{i\in J_\varepsilon(x)} \log\frac{2\varepsilon}{c_i(x)} \\ &= \alpha \sum_{i\in J_\varepsilon(x)} d_i + \sum_{i\in J_\varepsilon(x)} \log\frac{2\varepsilon r^{d_i}}{c_i(x)}. \end{aligned} \tag{12.2}$$

The empty sum is zero. Both summands in the last line are nonnegative. In particular this is not a signed logarithmic weight, and it introduces no clearance factor. The Borel witness set is allowed to vary with the roof and angle.

For a nonnegative Borel weight h, write

$$D[h](t) = \frac{dF_\#(h\mathbf{1}_\Pi \nu_R^*)}{dt}(t)$$

when this density is defined. The positive identities of the preceding section give a common version at fixed m, R, n, k. The next theorem also proves existence and finiteness for its weights.

Theorem 12.1 (Exponential moment of the original incidence load)**.** *For every finite collision count, every $\varepsilon > 0$ and $0 < \lambda < 1$,*

$$\left\|D[e^{\lambda \mathscr{L}_\varepsilon}]\right\|_\infty \le C_\star \exp\left\{\frac{4K_\star \varepsilon}{e^{\alpha(1-\lambda)} - 1}\right\}. \tag{12.3}$$

Consequently, for $u > b_\varepsilon$,

$$\left\|D[\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}}]\right\|_\infty \le C_\star \min\left\{1, \exp[-u + 2\sqrt{b_\varepsilon u}]\right\}. \tag{12.4}$$

Both estimates are uniform in the radius, exact labels and target roof. They remain valid with any further Borel restriction of the original measure, including the persistent source through Lemma 11.1.

Proof. Set $y_i = 2\varepsilon/c_i$. At a witness, $y_i > r^{-d_i}$, whence

$$y_i^\lambda = y_i y_i^{-(1-\lambda)} \le y_i r^{(1-\lambda)d_i}.$$

At every regular trajectory we therefore have the positive inequality

$$\begin{aligned} e^{\lambda \mathscr{L}_\varepsilon} &= \prod_{i=1}^{m-1} \left[1 + (y_i^\lambda - 1)\mathbf{1}_{\{i \in J_\varepsilon\}}\right] \\ &\le \prod_{i=1}^{m-1} \left(1 + \frac{2\varepsilon r^{(1-\lambda)d_i}}{c_i}\right). \end{aligned}$$

Apply the finite product insertion (10.6). There are at most two indices at every positive depth, so

$$\sum_{i=1}^{m-1} 2\varepsilon r^{(1-\lambda)d_i} \le 4\varepsilon \sum_{d \ge 1} r^{(1-\lambda)d} = \frac{4\varepsilon}{e^{\alpha(1-\lambda)} - 1}.$$

This proves (12.3) after the complete physical sum. Every product is finite before this last geometric upper bound is used; no independence of collisions is assumed.

Positivity gives $\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}} \le e^{-\lambda u} e^{\lambda \mathscr{L}_\varepsilon}$. Writing $v = 1 - \lambda$ and using $e^{\alpha v} - 1 \ge \alpha v$, we obtain, for every $0 < v < 1$,

$$\left\|D[\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}}]\right\|_\infty \le C_\star \exp\{-u + vu + b_\varepsilon / v\}.$$

When $u > b_\varepsilon$, take $v = \sqrt{b_\varepsilon / u}$. The independent bound $\|D[1]\|_\infty \le C_\star$ follows from the empty marked set and supplies the minimum in (12.4). Borel restrictions decrease a positive measure. For the persistent source the pointwise coefficient inequality in Lemma 11.1 supplies this decrease. □

Corollary 12.2 (Accumulated depth and severe shallow incidences)**.** *For $v > 0$, each of the events*

$$\left\{\alpha \sum_{i \in J_\varepsilon} d_i \ge v\right\}, \qquad \left\{\sum_{i \in J_\varepsilon} \log \frac{2\varepsilon r^{d_i}}{c_i} \ge v\right\}$$

is contained in $\{\mathscr{L}_\varepsilon \ge v\}$ and has the bound (12.4) whenever $v > b_\varepsilon$. An incidence witness satisfying $c_i \le 2\varepsilon e^{-v}$ is also in that event, irrespective of its depth.

Proof. Use the nonnegative decomposition in (12.2). For the last assertion, the single indicated summand is at least v. □

Thus a single sufficiently severe witness in an end layer can be paid. The conclusion does not require either a deep witness or the large witness count of Theorem 11.2. It does not assert a bound for an unvisited third-disk tangency.

13. Quadratic tails for the number of incidence witnesses

There is also an explicit finite-count improvement which retains all threshold locations, rather than passing to their total sum. The calculation is deterministic; it makes no independence claim about the billiard or its symbolic itinerary.

For $0 < r < 1$ write

$$(r;r)_a = \prod_{j=1}^{a}(1-r^j), \quad (r;r)_0 = 1, \quad (r;r)_\infty = \prod_{j=1}^{\infty}(1-r^j) > 0, \quad f(s) = \left\lfloor \frac{(s+1)^2}{4} \right\rfloor.$$

The infinite product is positive because the tail of $\sum_j -\log(1-r^j)$ is bounded by a constant times $\sum_j r^j$.

Lemma 13.1 (The two-end depth polynomial)**.** *For $0 \le s \le m-1$, put $E_{m,s}(r) = e_s((r^{d_i})_{1\le i<m})$. Then*

$$r^{f(s)} \le E_{m,s}(r) \le \sum_{a=0}^{s} \frac{r^{a(a+1)/2+(s-a)(s-a+1)/2}}{(r;r)_a (r;r)_{s-a}} \le \frac{(s+1)r^{f(s)}}{(r;r)_\infty^2}. \tag{13.1}$$

For $s > m-1$ the finite polynomial coefficient is zero and only the upper inequalities are asserted. In particular the coefficient has quadratic logarithmic rate, uniformly over $m \ge s+1$:

$$\log E_{m,s}(r) = -\frac{\alpha}{4}s^2 + O_r(s + \log(s+1)). \tag{13.2}$$

This is sharpness of the depth polynomial, not a lower bound for a physical roof density.

Proof. The multiset of finite collision depths is contained in two copies of $1, 2, 3, \ldots$. Coefficientwise positivity bounds its generating polynomial by $\prod_{d\ge1}(1+zr^d)^2$. To compute one factor, for $a \ge 1$ write $1 \le d_1 < \cdots < d_a$ as $d_j = j + h_j$, where $0 \le h_1 \le \cdots \le h_a$. Setting $g_1 = h_1$ and $g_j = h_j - h_{j-1}$ gives

$$\sum_{1\le d_1<\cdots<d_a} r^{d_1+\cdots+d_a} = r^{a(a+1)/2} \prod_{j=1}^{a} \sum_{g_j\ge0} r^{(a-j+1)g_j} = \frac{r^{a(a+1)/2}}{(r;r)_a}.$$

The empty factor contributes one. Multiplying the two positive series proves the middle sum in (13.1); all infinite passages are monotone limits of nonnegative coefficients. The minimum of its exponent over a is $f(s)$, attained by the two most balanced choices. Each denominator is at least $(r;r)_\infty^2$, giving the last inequality.

For the lower bound take the s smallest available depths. If $s = 2k$ these consist of two copies of $1, \ldots, k$ and have sum $k(k+1)$. If $s = 2k+1$ take these and one copy of $k+1$, with sum $(k+1)^2$. These depths are available whenever $m \ge s+1$, including the single central depth when m is even. Their product is one term of $E_{m,s}$. Taking logarithms of the two bounds proves (13.2). □

Theorem 13.2 (Quadratic witness-count tail at the target roof)**.** *With $N^\varepsilon = |J_\varepsilon|$, for every $s \ge 1$,*

$$\left\| D[\mathbf{1}_{\{N^\varepsilon \ge s\}}] \right\|_\infty \le \frac{C_\star (s+1)(2K_\star\varepsilon)^s}{(r;r)_\infty^2}\, r^{f(s)}. \tag{13.3}$$

The event is empty for $s > m-1$. The same estimate holds for the persistent source. For every fixed B and every $\eta > 0$, take

$$\sigma_m = \left\lceil 2\sqrt{\frac{(2+\eta)\log(m+1)}{\alpha}} \right\rceil. \tag{13.4}$$

Then the source with at least σ_m incidence witnesses has zero m^2-normalized essential height, with finite-count rate $m^{-\eta+o_B(1)}$.

Proof. Use the first, unrelaxed inequality in (10.8), with thresholds $2\varepsilon r^{d_i}$, and apply Lemma 13.1. Since $f(s) \geq s^2/4$, the logarithm of m^2 times the upper bound at $s = \sigma_m$ is at most

$$-\eta\log(m+1) + \sigma_m \log(2K_\star \varepsilon(B)) + \log(\sigma_m + 1) + \log \frac{C_\star}{(r;r)_\infty^2}.$$

For fixed B the last three terms are $O_B(\sqrt{\log(m+1)} + \log\log(m+2))$. This proves the claim. No outer width is changed. □

The square-root logarithmic threshold is strictly smaller in order than the logarithmic-over-logarithmic threshold of the earlier factorial bound. The estimate depends on the actual two-end depth geometry, not on additional cancellation or a count of physical words.

14. Positive payment and the remaining low-load source

We now apply the joint-load theorem to the original localized persistent source. All quantities $\varepsilon(B), h_B, r_B, \gamma_B, K_B$ are fixed at the outer stage. A load level used below classifies an original trajectory at finite collision count; it is not a spectral, roof, residence or comparison window.

Fix $\eta > 0$ and set

$$u_m = (2+\eta)\log(m+1), \qquad \mathcal{Q}_{B,m} = \{\mathscr{L}_{\varepsilon(B)} \geq u_m\}. \tag{14.1}$$

Keep the previously paid set $\mathcal{Z}_{B,m}$ from (11.6). Define two restrictions of the same positive source, with no new renormalization:

$$\begin{aligned} \mathsf{U}_B^{\mathrm{load}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c \cap \mathcal{Q}_{B,m}], \\ \mathsf{U}_B^{\mathrm{low}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c \cap \mathcal{Q}_{B,m}^c], \\ \mathsf{R}_B^{\mathrm{low}} &= (\mathsf{U}_B^{\mathrm{low}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{14.2}$$

The comparison pool is the full original nongrazing pool, not a source-selected pool. The notation low refers only to incidence load; clearance witnesses remain at arbitrary depth.

Theorem 14.1 (A new collision-normalized positive payment)**.** *At every finite count,*

$$\mathsf{U}_B^{\mathrm{per}} = \mathsf{U}_B^{\mathrm{exc}} + \mathsf{U}_B^{\mathrm{load}} + \mathsf{U}_B^{\mathrm{low}}, \qquad 0 \leq \mathsf{R}_B^{\mathrm{loc}} - \mathsf{R}_B^{\mathrm{low}} \leq \mathsf{U}_B^{\mathrm{load}}. \tag{14.3}$$

For every fixed B, M, the load term has zero ordered height:

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{load}}) = 0. \tag{14.4}$$

More explicitly, whenever $u_m > b_{\varepsilon(B)}$,

$$\sup_{R,n,k} m^2 \|\mathsf{U}_B^{\mathrm{load}}\|_\infty \leq C_\star \exp\left\{-\eta\log(m+1) + 2\sqrt{b_{\varepsilon(B)}(2+\eta)\log(m+1)}\right\}. \tag{14.5}$$

Consequently

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}} - \mathsf{R}_B^{\mathrm{low}}) = 0, \qquad \mathcal{E}_M \leq \mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}}) + C_M B^{-1/384}. \tag{14.6}$$

The original assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}}) \to 0$ for every fixed M.*

Proof. The three sets in (14.3) disjointly partition the original persistent source. Thus its first equality holds before pushforward and remains exact afterwards. Monotonicity and the unit Lipschitz bound of the positive-part map prove the residual inequality with the same capacity on both sides.

Lemma 11.1 and Theorem 12.1 bound $\mathsf{U}_B^{\mathrm{load}}$ by $C_\star \exp[-u_m + 2\sqrt{b_{\varepsilon(B)} u_m}]$ once $u_m > b_{\varepsilon(B)}$. Multiplication by m^2, followed by $m^2 \le (m+1)^2$, gives (14.5). At fixed B its exponent is $-\eta \log(m+1) + O_B(\sqrt{\log(m+1)})$, tending to minus infinity. The finitely many smaller counts do not affect the collision limsup. This proves (14.4) uniformly in all the stated inner suprema.

Apply the residual inequality and Corollary 11.3 to obtain (14.6). For necessity use $0 \le \mathsf{R}_B^{\mathrm{low}} \le \mathsf{U}_B^{\mathrm{micro}}$ and the original scalar equivalence. For sufficiency use (14.6) and then send B to infinity. This is the original order of limits. The optimizing parameter in the proof of (12.4) was chosen inside a finite-count inequality valid for every $0 < \lambda < 1$; it is not inserted into a theorem which holds only at a previously fixed outer width. □

Corollary 14.2 (Explicit support restrictions after payment)**.** *At any regular target point contributing to* $\mathsf{U}_B^{\mathrm{low}}$*, put* $J = J_{\varepsilon(B)}$ *and* $N = |J|$*. Then*

$$\alpha \sum_{i \in J} d_i < u_m, \qquad \alpha f(N) < u_m, \qquad \prod_{i \in J} \frac{c_i}{2\varepsilon(B)} > e^{-u_m}. \tag{14.7}$$

In particular $N < 2\sqrt{u_m/\alpha}$ *and every individual witness has* $c_i > 2\varepsilon(B) e^{-u_m}$*. All these restrictions are in addition to the retained v111 incidence localization. They impose no clearance localization.*

Proof. The load is strictly less than u_m on the retained set. Its nonnegative decomposition gives the first inequality. The sum of the depths of any N indices is at least $f(N)$, by the construction in Lemma 13.1; this proves the second. Exponentiating minus the definition of the load proves the product inequality. Since $f(N) \ge N^2/4$ and all load summands are nonnegative, the remaining assertions follow, including the empty-witness case. □

For sufficiently large B, the original section cap excludes incidence witnesses at the two endpoint collisions. The persistent guard then still forces an interior incidence witness or an original third-disk clearance witness. A state with $N = 0$ is not removed by this revision: the guard may be witnessed entirely by clearance. Critical entrances and section cuts carrying the original positive coefficient are likewise not deleted by their names.

The exact remaining target is

$$\lim_{B \to \infty} \limsup_{m \to \infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}_{B,m}^{\mathrm{low}}(t) - K_B P_{r_B,m}^{\mathrm{ng}}(t))_+ = 0. \tag{14.8}$$

Theorems 12.1–14.1 prove new finite-count concentration and positive payments in the required normalized topology. They do not prove (14.8). Low-load shallow incidences, arbitrary-depth clearance (including unvisited tangency), and capacity-deficient positive source remain in that equation. The original physical section, guard, arithmetic reference and exact-label pointwise assertion have not been weakened. The distinction between this remaining assertion and the proved payments is part of the theorem statement, not an inference from software diagnostics or an editorial change of topic.

Appendix A. Scope of the retained comparison statements

For clarity, we record the precise scope of the earlier allocation statements. None of their algebraic maximality assertions is used as a substitute for the physical estimate in Theorem 6.1.

Proposition A.1 (Fixed-window scalar maximality and old-family dominance)**.** *Fix a Borel comparison subspace, $0 < r < \Delta$, and $K > 0$. Consider nonnegative measurable weights w_+, w_- with $w_+ + w_- \le 1$, supported on the respective one-sided window conditions in* (5.2)*. Among scalar accepted heights $Z(t)$ satisfying*

$$0 \le Z(t) \le U(t), \qquad Z(t) \le K \int (w_+ + w_-) p_\vee \, d\theta,$$

the greatest is $V_{r,K}(t) = \min\{U(t), KP_r(t)\}$. The allocation is global at the roof and does not couple individual source sheets to comparison sheets.

For a compatible old fixed family with window width r_{old}, physical overlap N and block cost κ, the explicit conditions

$$0 < r \le r_{\mathrm{old}}, \qquad \ell_{\mathrm{comparison}} < \Delta, \qquad K = N\kappa$$

imply $V^{\mathrm{old}}(t) \le V_{r,N\kappa}(t)$ on the same original source and comparison space.

Proof. The support and one-copy conditions give $w_+ + w_- \le Q_r$. Hence every Z is bounded by the stated minimum. On $P_r > 0$ choose $w_\pm = V_{r,K} q_\pm / (KP_r)$; use zero weights otherwise. This is a measurable attaining certificate. The separate source multiplier is $\beta_{r,K}$ defined in Section 6. For the old family, retain only active source blocks at t. Their comparison supports lie in the r-pool by the displayed width condition. Summing the blockwise budget and using physical overlap at most N gives $V^{\mathrm{old}} \le N\kappa P_r$; disjoint original source blocks also give $V^{\mathrm{old}} \le U$. The minimum proves the assertion, with countable families obtained by monotone convergence on their common roof versions. □

Proposition A.2 (Strictness relative to roof-independent whole-interval blocks)**.** *In the interval-source model, the roofwise certificate class is strictly larger than the old class of roof-independent angular source blocks that requires a comparison interval to contain the translate of the* entire *source interval by $[0, r]$ or $[-r, 0]$. This statement compares certificate classes and does not assert a Lorentz realization.*

Proof. Let $s = 1/1000$ and $r = 1/10$. On two disjoint angular sets of measure one give both endpoint traces value one. Let their physical intervals be $(0, s)$ and $(3s/8, r + 7s/8)$, and select only the first as source. Take $\Delta > r + s/2$ and $K = 1$. For $3s/8 < t < 7s/8$ the second interval contains $[t, t + r]$, so $P_r(t) \ge U(t) = 1$. Neither physical interval has length $r + s$, which is necessary for the old translated-whole-source inclusion. Angular subdivision does not alter these intervals. Thus the old class cannot allocate a nonzero whole-source block while the roofwise class accepts the source on the indicated roof interval. □

A.1. **Limit order and the remaining estimate.** The constructions in this article are evaluated first at finite m, R, n, k with B, r_B, h_B, K_B already fixed. Positive physical sums are completed before taking the essential roof supremum, the radius and exact-label suprema, and the collision limsup. Only then is B sent to infinity. The assertion that every positive-length segment contains a sufficiently small window at a fixed count supplies no uniform modulus for this order.

The remaining estimate is precisely

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} \left(\mathsf{U}^{\mathrm{micro}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\right)_+ = 0.$$

No inference to this estimate is made from bounded wordwise optical weights, local integrability, a finite diagnostic, or a positive finite-count incidence. Theorem 6.1 proves the entire visited-grazing payment in this order; all remaining positive classes in the displayed expression are still part of the original pointwise problem.

Appendix B. Optical cofactors, multiple marks and simultaneous endpoints

We supply explicit matrix details for the optical identity and isolate the one-sided endpoint statement used in the trace theorem. These calculations take place on the original finite itinerary. They do not assert a bound for a sum over itineraries.

Lemma B.1 (The signed corner cofactor)**.** *For $m \geq 1$, with the empty determinant convention when $m = 1$,*

$$G_{0m} = (-1)^{m-1} \frac{R^2 c_0 c_m}{(\prod_{i=0}^{m-1} \tau_i) \det J}. \tag{B.1}$$

In particular the absolute value, not the signed mixed entry, gives $\omega = |G_{0m}|/R$.

Proof. Suppose first that $m > 1$ and put $n = m - 1$. The minor obtained by deleting the last row and first column of the tridiagonal matrix J is triangular, with diagonal entries $\tau_1^{-1}, \dots, \tau_{n-1}^{-1}$. Cramer's rule therefore gives

$$(J^{-1})_{1n} = (-1)^{n+1} \frac{\prod_{i=1}^{n-1} \tau_i^{-1}}{\det J}.$$

For $n = 1$ its numerator is the empty product and the formula says $(J^{-1})_{11} = 1/J_{11}$. The only endpoint–interior couplings are $H_{01} = R^2 c_0 c_1/\tau_0$ and $H_{nm} = R^2 c_n c_m/\tau_{m-1}$, whereas $H_{0m} = 0$. Since $H_{II}^{-1} = R^{-2} C_I^{-1} J^{-1} C_I^{-1}$, the Schur complement gives

$$G_{0m} = -\frac{R^2 c_0 c_m}{\tau_0 \tau_{m-1}} (J^{-1})_{1n},$$

which proves (B.1). If $m = 1$, the mixed entry follows directly from the single square in (3.2) and has positive sign.

For clarity, the equality between the determinant denominator and $\mathfrak{b}_m$ can also be checked without a sign convention for optical coordinates. Both quantities have initial values $B_0 = 0$, $B_1 = \tau_0$ and satisfy

$$B_{l+1} = \left(1 + \tau_l \kappa_l + \frac{\tau_l}{\tau_{l-1}}\right) B_l - \frac{\tau_l}{\tau_{l-1}} B_{l-1}.$$

For the determinant this is the tridiagonal recurrence after multiplication by the flight lengths. For the optical product it follows by eliminating the lower entry of its second column. This verifies the denominator identity used in Lemma 3.3. □

Put $T_{a,b} = \sum_{i=a}^{b-1} \tau_i$ for $0 \leq a < b \leq m$. For $S = \{j_1 < \cdots < j_s\} \subset \{1, \dots, m-1\}$ set $j_0 = 0$ and $j_{s+1} = m$.

Proposition B.2 (Positive expansion and simultaneous optical marks)**.** *The exact denominator has the positive expansion*

$$\mathfrak{b}_m = \sum_{S \subset \{1,\dots,m-1\}} \left(\prod_{j \in S} \kappa_j \right) \prod_{\nu=0}^{|S|} T_{j_\nu, j_{\nu+1}}. \tag{B.2}$$

For every fixed set S of s distinct interior marks,

$$\frac{\omega}{\prod_{j \in S} c_j} \leq \frac{R_+^{s+1}}{2^s \tau_-^{s+1}} \varrho^{m-s-1}. \tag{B.3}$$

For $S = \varnothing$ this is the first optical bound in (3.8)*; for $|S| = 1$ it is the second.*

Proof. Let $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Expand every factor $\mathsf{C}(\kappa) = I + \kappa E$ in the product (3.6). Between two selected marks the shears combine according to $\mathsf{S}(a)\mathsf{S}(b) = \mathsf{S}(a+b)$, and $E\mathsf{S}(T)E = TE$. The top-right entry of the term with marked set S is exactly the corresponding summand in (B.2). The empty term is $T_{0,m}$. Every coefficient is positive.

To obtain the stronger estimate retaining the unmarked reflections, expand only the factors at the prescribed marks. The coefficient of $\prod_{j\in S}\kappa_j$, with all other factors retained, is the product of the top-right entries of the $s+1$ intervening optical blocks. Indeed $E = e_2 e_1^{\mathsf{T}}$, so repeated multiplication factors that coefficient into these entries. The block with $j_{\nu+1} - j_\nu$ flights has top-right entry at least $\tau_- \varrho^{-(j_{\nu+1}-j_\nu-1)}$, by the same positive-column argument as in (3.9). Keeping only this nonnegative term gives

$$\mathfrak{b}_m \geq \left(\prod_{j\in S} \frac{2}{Rc_j}\right) \tau_-^{s+1} \varrho^{-(m-s-1)}.$$

Substitute this in $\omega = Rc_0c_m/\mathfrak{b}_m$, and use $c_0c_m \leq 1$ and $R \leq R_+$, to prove (B.3). □

This estimate controls simultaneous vanishing incidences on a fixed finite trajectory. Its constant depends on the number of marks, not on their locations. No word-count factor has been suppressed, and no pressure conclusion is drawn by summing its right side.

Lemma B.3 (One-sided limits at simultaneous physical mechanisms)**.** *Fix a positive-length pairing segment and approach one of its ends from the regular interior. The collision points, velocities, flight lengths, visited incidences and third-disk clearances have limits along the entire approach, even if several of the following occur simultaneously: visited grazing, unvisited tangency, a section boundary and a critical incoming endpoint. The limiting reduced Hessian and coarea factor are those in* (4.3) *and* (4.4)*. No continuation of the exact event through its boundary is asserted.*

Proof. Lemma 4.1 gives limits for q and z along the entire segment, not only a subsequence. Hence the initial collision point and $p_0 = -z\cos\theta/R$ converge. The section cap gives the outgoing normal component $c_0 > 0$ at the limit, so the initial velocity converges as well.

Suppose inductively that the departure point x and unit velocity v of a designated flight converge. For its designated next disk with centre O, the entry time is

$$\tau = -(x-O)\cdot v - \sqrt{((x-O)\cdot v)^2 - |x-O|^2 + R^2}.$$

This is the entry root on the actual regular itinerary. Its radicand is nonnegative along the approach and has a nonnegative limit. The root is continuous also when the radicand tends to zero; $\tau \geq \tau_- > 0$ prevents a zero-length flight. The arrival point therefore converges. Reflection by $v \mapsto v - 2(v\cdot n)n$ is continuous even at tangency, so the next velocity converges. Induction applies a finite number of times. An earlier disk may become tangent only at the limiting flight; this does not alter the designated entry-root limit taken from the original event.

At fixed itinerary and radius, the bounded flight lengths restrict all possibly nearby third-disk centres to a finite set. Distance from a compact line segment to each such disk is continuous in the endpoints. Their finite minimum, and hence the original clearance, has a limit. A section decision may change at the limiting state, but its coordinate limit is still obtained from the same side. None of these continuity arguments requires the boundary mechanisms to occur separately. Formula (4.3), with Z^{-1} uniformly bounded, then gives the Hessian limit even with several vanishing incidences. The fixed-direction coarea formula gives the asserted limit for χ. □

Corollary B.4 (An explicit uniform trace modulus)**.** *A permissible constant in* (4.6) *is*

$$C_{\mathrm{H}} = \frac{500 C_{\mathrm{reg}} \sqrt{2\Lambda_\circ}}{4\pi c_* \lambda_\circ^2}. \tag{B.4}$$

It is common to all original sheets, collision counts, radii and exact labels. Its inherited dependence is precisely the regular variation constant C_{reg}; the remaining constants are the fixed geometry and section constants displayed in this article.

Proof. Corollary 3.4 and $\chi \geq \lambda_\circ$ give $AI_m \leq 500\varrho^{(m-2)_+}/(4\pi c_* \lambda_\circ)$. Integrate the regular derivative bound between u and v, and apply Lemma 4.1 to that subsegment:

$$\int_u^v z^{-1}\, ds \leq \lambda_\circ^{-1} \sqrt{2\Lambda_\circ (v-u)}.$$

Multiply the constants and pass to the already proved endpoints. □

Any nonempty collection of visited endpoint grazings belongs to Γ, and one vanishing marked incidence is already sufficient for its zero source statement. Proposition B.2 additionally controls their simultaneous product. This does not alter the distinct treatment of an unvisited third-disk tangency.

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